

Supplementary Material

This document is the **Supplementary Material of the manuscript**, submitted together with it.

Numbering convention: the main text uses §n for sections and S n for subsections of this document.

0. Generation and verifiability (**read this section first**)

This document **was not written by hand**: the body text of every section **embeds verbatim** the content of the following **verified source files**,

so "the numbers in this document" and "the numbers in the source files" **cannot be inconsistent**. Every source file records an md5 and a byte count, so anyone can re-check it.

Subsection	Content	Source file	md5	Bytes
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S1	S1 · Per-row marking table and mechanical recomputation over 19 public benchmarks × 4 counting units	split_units_19rows_20260916.md	2d542d10226347dffffb69dbab8b61d1a	5805
S2	S2 · Correction record for the counting convention (old convention → recomputed convention)	the audit-correction and adjudication record, §2 superseding block	94e85a41a6f7c5a5b6788a11b96c2c8e	24638
S3	S3 · Per-row evidence from the cross-framework configuration audit (5 frameworks / 23 official files)	xframe_evidence_20260916.txt	78b3e251f5d4c8db3b6378ecbbd2b3e0	13361
S4	S4 · 2×2 evaluation matrix over the three-split protocol (41 runs × {best,last} × {val,test})	xeval_analysis_20260916.txt	8b74b2eb6228c078d5a38131d119567b	8534

S5	S5 · Selection premium on public third-party logs (sample and how to read it)	selection_premium_public_20260916.md	04950c92d1ab4bcfb5bbe616b2dcbec4	9479
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Generation script: work\gen_supplementary_20260916.py (purely local and re-runnable; S5 reads the collaborating archive in read-only mode).

Re-check entry point: see §S7 of this document.

Terms this paper uses (one place, so they need not be reconstructed)

term	what it means here
---	---
arm	one member of a paired comparison. The baseline arm follows the published convention; the strategy arm is the intervention under test (e.g. a learning-rate change). "Baseline" does not mean unmodified software — both arms run the same code
cell	one (corpus pair × arm) unit of the design. A cell name such as shwd2sf reads source → target (SHWD → SFCHD); the corpora are dota15, aitod, visdrone, smoke, shwd, sfchd, mende, mask
`best` / `last`	the two checkpoints compared throughout: best.pt , written by the training script by fitness on the monitored split, and the final epoch
selection premium	best – last within a run : what selecting the best checkpoint adds, measured on the split the numbers are reported from
Δgap	the paired difference of the premium between the two arms , per corpus. Positive means val understates the strategy arm's effect (§5.1, Tier 2)
realization rate	prem_test / prem_val , aggregated over cells (§8.6); the share of the premium selected on val that survives on a disjoint test
κ	through-origin slope of ΔT on ΔV over a run's epochs (§8.7): how much of a run's movement on val appears on test
same directory / alias	a dataset config whose val: and test: point at the same path, so the checkpoint is selected on the split it is scored on
`shapeiou`	the shape-aware IoU variant used in the registered replication's strategy arm (§9)
`sio_b_results.csv`	the per-run results table of the clean-protocol batch; its unwritten endpoint convention is disclosed as a defect in §10
FROZEN-HASH	the registration's self-referential stamp: the md5 of the registration text above the stamp line (§9, verifiable by re-hashing that text)

S1 · Per-row marking table and mechanical recomputation over 19 public benchmarks × 4 counting units

Purpose: main text §3, §6; this is the **complete per-row evidence** for the claim "how many benchmarks are problematic".

Source: split_units_19rows_20260916.md (md5 2d542d10226347dffffb69dbab8b61d1a, 5805 bytes); the content below is **embedded verbatim, unmodified**.

19-row × 5-column unit marking table (front matter of the companion paper)

> Generated by work/recompute_split_units_20260916.py; all percentages are mechanically recomputed from the markings in this table, with nothing hard-coded.

> Established 2026-09-16. The (h-1) ruling has been applied (D-Fire counts as clean).

The three units (the main text must state which one it uses)

Unit	Definition
release	The benchmark's own release layer: whether the official split artifacts/scripts provide an independent held-out test
protocol	The official protocol layer: the split declared by the official source, and whether that test can be self-evaluated locally
yolo_dist	Generic YOLO distribution-package layer (= the column measured in the §1 table of the audit, i.e. column 4)
reported	The actually-reported layer: which split the numbers reported in the literature come from, and whether they are independent of checkpoint selection

Evidence marks: (A) explicit in the audit | (D) derived from audit facts | (?) the audit takes no position

Marking table

#	benchmark	release	protocol	yolo_dist	reported	Evidence category	Confidence
1	COCO	n_a(A)	independent_test(A)	clean(A)	clean(A)	Official counts + mirror yaml	high
2	PASCAL VOC	n_a(A)	test_gated(A)	alias(A)	alias(A)	mirror VOC.yaml (both keys cited per row)	high
3	Objects365	no_test(A)	no_test(A)	no_test_key(A)	alias(D)		

```

mirror yaml + Ultralytics docs | high |
| 4 | Open Images v7 | independent_test(A) | independent_test(A) |
no_test_key(A) | alias(D) | Official facts page + mirror yaml | high |
| 5 | DOTA v1.0 | independent_test(A) | independent_test(A) | clean(A) |
clean(A) | Official download page + mirror DOTAv1.yaml | high |
| 6 | DOTA v2.0 | independent_test(A) | test_gated(A) | clean(A) | alias(D) |
Official/mirror documentation | high |
| 7 | VisDrone-DET | independent_test(A) | independent_test(A) | clean(A) |
clean(A) | mirror + converter | high |
| 8 | AI-TOD | contradictory(A) | contradictory(A) | unknown(A) | unknown(?) |
Official README (contains two mutually contradictory release statements)
| low |
| 9 | UAVDT | no_val(A) | no_val(A) | unknown(A) | alias(A) | Official paper
+ two mirrors (the audit calls these unverified beyond these two) |
medium |
| 10 | xView | no_test(A) | no_test(A) | no_test_key(A) | alias(A) | mirror
xView.yaml | low |
| 11 | DIOR | independent_test(A) | independent_test(A) | no_test_key(A) |
unknown(?) | Official description (no counting) + mirror DIOR-R yaml |
low |
| 12 | NWPU VHR-10 | no_split(A) | no_split(A) | unknown(A) | alias(A) |
Official TorchGeo (the split means positive/negative) | high |
| 13 | SHWD | no_yaml(A) | no_val(A) | train_val_alias(A) | alias(A) |
Official README + mirror Reflective_vests.yaml | medium |
| 14 | SFCHD | alias(A) | no_val(A) | alias(A) | alias(A) | Official first-
party release (machine comparison _ev_sfchd_check2.py) | highest |
| 15 | MAFA | no_val(A) | no_val(A) | unknown(?) | alias(A) | Official CVPR
PDF (extracted with pypdf) | high |
| 16 | Mendeley face-mask | no_split(A) | no_split(A) | alias(D) | unknown(?)
| Official data record + mirror generator | lowest |
| 17 | WIDER FACE | independent_test(A) | test_gated(A) | no_test_key(A) |
alias(A) | Official site (GT not released, verified word by word) +
mirror | high |
| 18 | CrowdHuman | test_gated(A) | alias(A) | no_test_key(A) | alias(A) |
Official paper + mirror | high |
| 19 | D-Fire (FireSmoke) | independent_test(A) | independent_test(A) |
clean(A) | unknown(?) | Official README | medium |

```

Recomputation results

```

| Unit | alias/non-independent | Among decidable rows | unknown |
|---|---|---|---|
| release (the benchmark's own release layer) | 10/19 = 52.6% | 10/19 =

```

52.6% | 0 |
 | protocol (official protocol layer) | **13/19 = 68.4%** | 13/19 = 68.4% |
 0 |
 | yolo (generic YOLO distribution-package layer (yolo_dist)) | **4/19 = 21.1%** | 4/15 = 26.7% | 4 |
 | reported (actually-reported layer) | **12/19 = 63.2%** | 12/15 = 80.0% |
 4 |

Cross-check against the numbers already published in the audit

- Audit 3a "literal val == test **2/19**" → this table's "the two keys of the distribution-package config point to the same path" (A evidence source) yields **2/19, reproduced ✓**
 - The other 1 alias row is produced by a **generator** (Mendeley face-mask), not by a distribution-package config, and is not counted under the 3a convention
- Audit "**14/19**" → this table's reported yields **12/19** (difference -2)
- Audit "**5/19 clean**" → this table's reported=clean yields **3/19** (difference -2)
- The audit's "12/19" was obtained by manually adjusting two rows and is **not** a recomputation under a consistent unit; this table recomputes by a consistent rule, so the values need not be identical

A self-consistent set of numbers for the main text to adopt

- Literal val == test (generic distribution-package config, A evidence source): **2/19 = 11%** – PASCAL VOC, SFCHD
- train/val from the same source, produced by the generator: **1/19** (Mendeley face-mask), **a different convention, which must be stated separately**
- Non-independent at the official protocol layer: **13/19 = 68%**
- Non-independent at the actually-reported layer: **12/19 = 63%** (4 unknown rows) – **13/19** if a
 - paper's own val numbers, rather than community test-dev numbers, are what the unit judges
 - for COCO (the main text states both readings)
- Clean at the actually-reported layer: **3/19 = 16%**

The four levels of "same directory" (resolving (g0))

- **L1** literal val==test in a generic distribution package: 2 rows – PASCAL VOC, SFCHD
- **L2** distribution package with train==val (F-C): 1 row – SHWD

- **L3** distribution package clean but reported layer non-independent: 9 rows – Objects365, Open Images v7, DOTA v2.0, UAVDT, xView, NWPU VHR-10, MAFA, WIDER FACE, CrowdHuman
- **L4** all the rest: 7 rows

Corrections already applied

- **(h-1) D-Fire / FireSmoke family:** the "mixed" criterion for the family row is taken from a config of SHWD lineage (A:59 cites Reflective_vests.yaml; A:52 notes that it is based on SHWD). All three copies belonging to FireSmoke/D-Fire are distinct. Under the "release layer" unit explicitly adopted in 3a, D-Fire goes into the clean group.
- **(d-1) AI-TOD / DOTA asymmetry:** the audit marks F-C for AI-TOD, which contains trainval, but does not mark DOTA v1/v2, which likewise contain trainval. This table uniformly adopts the "release/protocol layer" unit: trainval is a user-protocol caveat and is not used as grounds for a verdict; neither place is re-judged, but AI-TOD is contradictory because its v2 release statement contradicts itself.

S2 · Correction record for the counting conventions (old convention → recomputed convention)

Purpose: main text §6.3; explains why the same fact once had two sets of numbers, and which set has been invalidated.

Source: the audit-correction and adjudication record, §2 superseding block (md5 94e85a41a6f7c5a5b6788a11b96c2c8e, 24638 bytes); the following content is **embedded verbatim, unmodified**.

```
> ### The four counts in this section (§2) have been superseded by a
mechanical recomputation (note added 2026-09-16)
>
> The 12/19 (release/protocol layer), 13/19 (actual reporting layer),
6/19 (clean) and ≤14/19 (upper bound) in the §2 table
> are inconsistent with the mechanical recomputation completed 4
minutes later, and must no longer be cited. The recomputation is the
sole counting source: analysis/eval_validity/split_units_recompute_20260916.txt
and
> split_units_19rows_20260916.md
>
> | Unit | Non-independent | Clean | Undecidable |
> |---|---|---|---|
> | Literal val == test (the benchmark's own release) | 2/19 ≈ 11% | – |
```

– |

> | release (release layer) | **10/19 ≈ 53%** | – | – |

> | protocol (official protocol layer) | **13/19 ≈ 68%** | – | – |

> | yolo_dist (generic YOLO distribution package layer) | **4/19 ≈ 21%** |

– | 4 |

> | **reported (actual reporting layer)** | **12/19 ≈ 63%** (12/15 = **80%** of the decidable rows) | **3/19 ≈ 16%** | 4 |

>

> The four "undecidable" rows are AI-TOD, DIOR, Mendeley and D-Fire – **the audit expressed no position on their reporting practice**; it does not mean they are clean.

>

> The four concrete discrepancies and their dispositions:

> 1. §2's "release/protocol layer 12/19" does not hold under the recomputed convention – the recomputed release is 10/19 and protocol is 13/19,

> **12 is neither the one nor the other** (consistent with the ruling in §1(g): it is the result of two rows being hand-adjusted).

> 2. §2's "actual reporting layer 13/19" should be **12/19** (4 rows undecidable).

> 3. §2's "clean 6/19" should be **3/19**; the reason the clean group dropped from 5 to 3 is that those 4 rows **were never addressed by the audit** – this is a finding, not a gap.

> 4. §2's "≤14/19 upper bound" is replaced by a derivable statement: **80% of the decidable rows are non-independent (12/15)**, which is both stronger and more defensible than an unenumerated upper bound.

>

> **"79%" has already been ruled inadmissible in §1(e)**; this is confirmed here as well: the main text uses only the four enumerated points in the table above.

> **⚠ This section was revised on 2026-09-16 (second version). The first version (09:56) was written before the 19×5 marking table was built and gave predicted values**

> (release/protocol layer 12/19, actual reporting layer 13/19, clean 6/19). The mechanical recomputation 4 minutes later

> (eval_validity\split_units_recompute_20260916.txt, 10:00, with a per-unit 19-row subtotal check) gave different results.

> **The two sets of numbers cannot be cited side by side.** Following the disposition recommendation in §7 #0 of the asset inventory,

> **the recomputation is taken as the sole counting source**; the first version's predicted values are retained in the "trail" table below for traceability.

S3 · Line-by-line evidence for the cross-framework configuration audit (5 frameworks / 23 official files)

Purpose: main text §6.5; the commit pin and the line-by-line evidence for each framework (the judgements are in the main text, the evidence is here).

Source: xframe_evidence_20260916.txt (md5 78b3e251f5d4c8db3b6378ecbbd2b3e0, 13361 bytes); the following content is **embedded verbatim, unmodified**.

```
=====
=====
```

Item C: extraction of cross-framework "split and checkpoint-selection" evidence (mechanical extraction, with line numbers; no judgements in this file)

```
=====
=====
```

```
repo      default_branch  commit_sha
mmdetection  open-mmlab/mmdetection  main
           cfd5d3a985b0249de009b67d04f37263e11cdf3d
detectron2   facebookresearch/detectron2  main
           a2f4a8771ab77e8411c26b27f24f9489a28a2453
yolox       Megvii-BaseDetection/YOLOX    main
           6ddff4824372906469a7fae2dc3206c7aa4bbaee
paddledetection  PaddlePaddle/PaddleDetection  release/2.9
           b25522a0f4bde8c80603f3ba5e3472059972e3b5
detr        facebookresearch/detr    main
           29901c51d7fe8712168b8d0d64351170bc0f83e0
```

```
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-----
```

MMDetection ::

mmdetection/configs__base__datasets__coco_detection.py (96 lines)

```
3: data_root = 'data/coco/'
21: dict(type='LoadImageFromFile', backend_args=backend_args),
22: dict(type='LoadAnnotations', with_bbox=True),
23: dict(type='Resize', scale=(1333, 800), keep_ratio=True),
24: dict(type='RandomFlip', prob=0.5),
25: dict(type='PackDetInputs')
28: dict(type='LoadImageFromFile', backend_args=backend_args),
29: dict(type='Resize', scale=(1333, 800), keep_ratio=True),
31: dict(type='LoadAnnotations', with_bbox=True),
```

```

33:         type='PackDetInputs',
41:     sampler=dict(type='DefaultSampler', shuffle=True),
42:     batch_sampler=dict(type='AspectRatioBatchSampler'),
45:     data_root=data_root,
46:     ann_file='annotations/instances_train2017.json',
47:     data_prefix=dict(img='train2017/'),
56:     sampler=dict(type='DefaultSampler', shuffle=False),
59:     data_root=data_root,
60:     ann_file='annotations/instances_val2017.json',
61:     data_prefix=dict(img='val2017/'),
68:     type='CocoMetric',
69:     ann_file=data_root + 'annotations/instances_val2017.json',

```

```

-----
##### MMDetection :: mmdetection/configs__base__default_runtime.py
(25 lines)
(no hits)

```

```

-----
##### MMDetection :: mmdetection/configs__faster_rcnn__faster-
rcnn_r50_fpn_1x_coco.py (6 lines)
1: _base_ = [
2:     '../_base_/models/faster-rcnn_r50_fpn.py',
3:     '../_base_/datasets/coco_detection.py',
4:     '../_base_/schedules/schedule_1x.py',
'../_base_/default_runtime.py'
5: ]

```

```

-----
##### MMDetection :: mmdetection/tools__train.py (122 lines)
(no hits)

```

```

-----
##### Detectron2 :: detectron2/configs__COCO-
Detection__faster_rcnn_R_50_FPN_1x.yaml (7 lines)
1: _BASE_: "../Base-RCNN-FPN.yaml"
2: MODEL:
3:     WEIGHTS: "detectron2://ImageNetPretrained/MSRA/R-50.pkl"
4:     MASK_ON: False

```

```
5:   RESNETS:
6:     DEPTH: 50
```

```
-----
##### Detectron2 :: detectron2/configs__Base-RCNN-FPN.yaml   (43 lines)
14:   PRE_NMS_TOPK_TRAIN: 2000 # Per FPN level
15:   PRE_NMS_TOPK_TEST: 1000  # Per FPN level
19:   POST_NMS_TOPK_TRAIN: 1000
20:   POST_NMS_TOPK_TEST: 1000
32: DATASETS:
33:   TRAIN: ("coco_2017_train",)
34:   TEST: ("coco_2017_val",)
36:   IMS_PER_BATCH: 16
41:   MIN_SIZE_TRAIN: (640, 672, 704, 736, 768, 800)
```

```
-----
##### Detectron2 :: detectron2/tools__train_net.py   (168 lines)
105:           for name in cfg.DATASETS.TEST
```

```
-----
##### Detectron2 :: detectron2/detectron2__data__datasets__builtin.py
(260 lines)
11:   without having to download the dataset annotations
36:   "coco_2014_train": ("coco/train2014",
"coco/annotations/instances_train2014.json"),
37:   "coco_2014_val": ("coco/val2014",
"coco/annotations/instances_val2014.json"),
38:   "coco_2014_minival": ("coco/val2014",
"coco/annotations/instances_minival2014.json"),
41:   "coco/annotations/instances_valminusminival2014.json",
43:   "coco_2017_train": ("coco/train2017",
"coco/annotations/instances_train2017.json"),
44:   "coco_2017_val": ("coco/val2017",
"coco/annotations/instances_val2017.json"),
45:   "coco_2017_test": ("coco/test2017",
"coco/annotations/image_info_test2017.json"),
46:   "coco_2017_test-dev": ("coco/test2017",
"coco/annotations/image_info_test-dev2017.json"),
47:   "coco_2017_val_100": ("coco/val2017",
"coco/annotations/instances_val2017_100.json"),
```

```

53:         "coco/annotations/person_keypoints_train2014.json",
55:         "keypoints_coco_2014_val": ("coco/val2014",
"coco/annotations/person_keypoints_val2014.json"),
58:         "coco/annotations/person_keypoints_minival2014.json",
62:
"coco/annotations/person_keypoints_valminusminival2014.json",
64:         "keypoints_coco_2017_train": (
66:             "coco/annotations/person_keypoints_train2017.json",
68:             "keypoints_coco_2017_val": ("coco/val2017",
"coco/annotations/person_keypoints_val2017.json"),
69:             "keypoints_coco_2017_val_100": (
70:                 "coco/val2017",
71:                 "coco/annotations/person_keypoints_val2017_100.json",
77:             "coco_2017_train_panoptic": (
80:                 "coco/annotations/panoptic_train2017.json",
88:             "coco_2017_val_panoptic": (
89:                 "coco/panoptic_val2017",
90:                 "coco/annotations/panoptic_val2017.json",
91:                 "coco/panoptic_stuff_val2017",
93:             "coco_2017_val_100_panoptic": (
94:                 "coco/panoptic_val2017_100",
95:                 "coco/annotations/panoptic_val2017_100.json",
96:                 "coco/panoptic_stuff_val2017_100",
235:         gt_dir = os.path.join(root, "annotations_detectron2",
dirname)

```

```

-----
### Detectron2 :: detectron2/detectron2__config__defaults.py    (657
lines)

```

```

103: _C.DATASETS.TEST = ()
607: _C.TEST.EVAL_PERIOD = 0

```

```

-----
### YOLOX :: yolox/exps__default__yolox_s.py    (16 lines)

```

```

7: from yolox.exp import Exp as MyExp
15:         self.exp_name =
os.path.split(os.path.realpath(__file__))[1].split(".")[0]

```

```

-----
### YOLOX :: yolox/yolox__data__datasets__coco.py    (189 lines)

```

```
42:         json_file="instances_train2017.json",
43:         name="train2017",
54:         name (str): COCO data name (e.g. 'train2017' or
'val2017')
```

```
-----
##### YOLOX :: yolox/tools__train.py    (147 lines)
(no hits)
```

```
-----
##### YOLOX :: yolox/tools__eval.py    (221 lines)
93:         "--test",
```

```
-----
##### PaddleDetection ::
paddledetection/configs__yolov3__yolov3_darknet53_270e_coco.yml    (11
lines)
1: _BASE_: [
2:   '../datasets/coco_detection.yml',
3:   '../runtime.yml',
4:   '_base_/optimizer_270e.yml',
5:   '_base_/yolov3_darknet53.yml',
6:   '_base_/yolov3_reader.yml',
7: ]
9: snapshot_epoch: 5
10: weights: output/yolov3_darknet53_270e_coco/model_final
```

```
-----
##### PaddleDetection :: paddledetection/tools__train.py    (220 lines)
51:         "--eval",
54:         help="Whether to perform evaluation in train")
66:         help="If set True, enable continuous evaluation job."
94:         help='Whether to save the evaluation results only')
100:         "format \"key1=value1;key2=value2;key3=value3\"."
161:     trainer.train(FLAGS.eval)
```

```
-----
##### DETR :: detr/main.py    (249 lines)
```

```

    16: from engine import evaluate, train_one_epoch
    95:     parser.add_argument('--eval', action='store_true')
   143:     dataset_val = build_dataset(image_set='val', args=args)
   147:     sampler_val = DistributedSampler(dataset_val,
shuffle=False)
    150:     sampler_val =
torch.utils.data.SequentialSampler(dataset_val)
    157:     data_loader_val = DataLoader(dataset_val, args.batch_size,
sampler=sampler_val,
    162:     coco_val = datasets.coco.build("val", args)
    163:     base_ds = get_coco_api_from_dataset(coco_val)
    165:     base_ds = get_coco_api_from_dataset(dataset_val)
    179:     if not args.eval and 'optimizer' in checkpoint and
'lr_scheduler' in checkpoint and 'epoch' in checkpoint:
    184:         if args.eval:
    185:             test_stats, coco_evaluator = evaluate(model, criterion,
postprocessors,
    186:                                                     data_loader_val,
base_ds, device, args.output_dir)
    188:     utils.save_on_master(coco_evaluator.coco_eval["bbox"].eval, output_dir
/ "eval.pth")
    214:     test_stats, coco_evaluator = evaluate(
    215:         model, criterion, postprocessors, data_loader_val,
base_ds, device, args.output_dir
    228:         if coco_evaluator is not None:
    229:             (output_dir / 'eval').mkdir(exist_ok=True)
    230:             if "bbox" in coco_evaluator.coco_eval:
    235:
torch.save(coco_evaluator.coco_eval["bbox"].eval,
    236:         output_dir / "eval" / name)
    244:     parser = argparse.ArgumentParser('DETR training and
evaluation script', parents=[get_args_parser()])

```

```

-----
### DETR :: detr/datasets__coco.py    (159 lines)

```

```

    151:     PATHS = {
    152:         "train": (root / "train2017", root / "annotations" /
f'{mode}_train2017.json'),
    153:         "val": (root / "val2017", root / "annotations" /
f'{mode}_val2017.json'),
    156:     img_folder, ann_file = PATHS[image_set]

```

```

-----
##### DETR :: detr/engine.py (152 lines)
    3: Train and eval functions used in main.py
    13: from datasets.coco_eval import CocoEvaluator
    14: from datasets.panoptic_eval import PanopticEvaluator
    23:     metric_logger.add_meter('lr',
utils.SmoothedValue(window_size=1, fmt='{value:.6f}'))
    24:     metric_logger.add_meter('class_error',
utils.SmoothedValue(window_size=1, fmt='{value:.2f}'))
    43:     losses_reduced_scaled =
sum(loss_dict_reduced_scaled.values())
    45:     loss_value = losses_reduced_scaled.item()
    47:     if not math.isfinite(loss_value):
    48:         print("Loss is {}, stopping
training".format(loss_value))
    58:     metric_logger.update(loss=loss_value,
loss_dict_reduced_scaled, loss_dict_reduced_unscaled)
    68: def evaluate(model, criterion, postprocessors, data_loader,
base_ds, device, output_dir):
    69:     model.eval()
    70:     criterion.eval()
    73:     metric_logger.add_meter('class_error',
utils.SmoothedValue(window_size=1, fmt='{value:.2f}'))
    77:     coco_evaluator = CocoEvaluator(base_ds, iou_types)
    80:     panoptic_evaluator = None
    82:     panoptic_evaluator = PanopticEvaluator(
    85:         output_dir=os.path.join(output_dir,
"panoptic_eval"),
    102:
metric_logger.update(loss=sum(loss_dict_reduced_scaled.values()),
    113:         if coco_evaluator is not None:
    114:             coco_evaluator.update(res)
    116:         if panoptic_evaluator is not None:
    124:             panoptic_evaluator.update(res_pano)
    129:     if coco_evaluator is not None:
    130:         coco_evaluator.synchronize_between_processes()
    131:     if panoptic_evaluator is not None:
    132:         panoptic_evaluator.synchronize_between_processes()
    135:     if coco_evaluator is not None:
    136:         coco_evaluator.accumulate()
    137:         coco_evaluator.summarize()

```

```

139:     if panoptic_evaluator is not None:
140:         panoptic_res = panoptic_evaluator.summarize()
142:     if coco_evaluator is not None:
144:         stats['coco_eval_bbox'] =
coco_evaluator.coco_eval['bbox'].stats.tolist()
146:         stats['coco_eval_masks'] =
coco_evaluator.coco_eval['segm'].stats.tolist()
151:     return stats, coco_evaluator

```

```

=====
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```

Missing files: none

S4 · 2×2 evaluation matrix over the three-split protocol (41 runs × {best,last} × {val,test})

Purpose: main text §8.6 and §5.6; covers archive-reproduction verification, the decomposability of Δ_{gap} , and the realization rate of the selection gain.

Source: xeval_analysis_20260916.txt (md5 8b74b2eb6228c078d5a38131d119567b, 8534 bytes); the content below is **embedded verbatim, unchanged**.

```

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```

2×2 evaluation matrix × archive: verification / locally recomputable / decomposability of Δ_{gap}

```

=====
=====

```

Matrix entries 164; sio entries 472; (commit/convention: split=test|val, batch=32, imgsz=640, aligned with train_obj.py L75)
Multiple evaluations of the same cell: all consistent

--- per run: archive vs this recomputation (pp)

run	Vmax_a	Vbest_r	Vfin_a
Vfin_r Tbest_a Tbest_r Tfin_r			
r10_aitod20_base30_3way_s42n	14.621	14.621	13.835
13.826 10.210 10.211 10.587			
r10_shwd2sf_base100_3way_s42n	60.694	60.707	59.055
58.977 43.510 43.513 42.946			
r10_shwd2sf_base100_3way_s43n	60.976	60.966	59.761
59.822 43.880 43.885 42.948			
r10_shwd2sf_base100_3way_s44n	61.120	61.135	59.230
59.326 43.050 43.053 42.394			
r10_shwd2sf_base100_3way_s45n	59.852	59.866	58.473

58.474	43.080	43.079	42.901			
r10_shwd2sf_base100_3way_s46n				60.509	60.507	59.293
59.284	43.640	43.641	42.845			
r10_shwd2sf_base100_3way_s47n				60.858	60.886	59.766
59.775	43.780	43.781	43.267			
r10_shwd2sf_base100_3way_s48n				60.554	60.516	59.575
59.615	43.390	43.390	42.784			
r10_shwd2sf_base100_3way_s49n				61.026	60.999	59.169
59.248	43.990	43.992	42.910			
r10_shwd2sf_base100_3way_s50n				60.896	60.854	58.967
59.018	43.640	43.641	42.608			
r10_shwd2sf_base100_3way_s51n				60.209	60.208	58.713
58.734	43.630	43.626	42.636			
r10_shwd2sf_lr005_100ep_3way_s42n				60.860	60.913	59.581
59.586	44.260	44.260	43.569			
r10_shwd2sf_lr005_100ep_3way_s43n				60.408	60.491	59.116
59.105	43.950	43.947	43.861			
r10_shwd2sf_lr005_100ep_3way_s44n				60.001	60.003	58.952
58.920	44.440	44.442	44.134			
r10_shwd2sf_lr005_100ep_3way_s45n				60.795	60.789	60.578
60.579	44.130	44.132	43.934			
r10_shwd2sf_lr005_100ep_3way_s46n				60.462	60.456	59.270
59.298	44.350	44.354	44.242			
r10_shwd2sf_lr005_100ep_3way_s47n				60.678	60.667	60.354
60.355	43.970	43.966	43.944			
r10_shwd2sf_lr005_100ep_3way_s48n				59.722	59.742	58.701
58.711	43.710	43.713	43.795			
r10_shwd2sf_lr005_100ep_3way_s49n				60.363	60.375	59.037
59.028	43.800	43.799	44.027			
r10_shwd2sf_lr005_100ep_3way_s50n				60.193	60.188	59.614
59.634	43.780	43.785	44.078			
r10_shwd2sf_lr005_100ep_3way_s51n				60.491	60.490	60.119
60.153	44.270	44.269	43.560			
r10_smoke2sf_base100_3way_s42n				58.724	58.696	57.618
57.689	42.100	42.103	41.661			
r10_smoke2sf_base100_3way_s43n				58.766	58.798	58.438
58.449	42.320	42.323	41.726			
r10_smoke2sf_base100_3way_s44n				58.864	58.882	58.513
58.526	42.280	42.282	41.367			
r10_smoke2sf_base100_3way_s45n				58.731	58.740	57.594
57.592	41.250	41.253	41.054			
r10_smoke2sf_base100_3way_s46n				58.652	58.689	57.887
57.862	41.900	41.899	41.258			

r10_smoke2sf_base100_3way_s47n	59.271	59.228	58.532
58.557 42.100 42.101 41.437			
r10_smoke2sf_base100_3way_s48n	58.697	58.682	57.778
57.755 42.400 42.402 41.441			
r10_smoke2sf_base100_3way_s49n	58.601	58.617	58.376
58.393 41.960 41.964 42.082			
r10_smoke2sf_base100_3way_s50n	58.449	58.444	57.430
57.410 42.110 42.108 41.923			
r10_smoke2sf_base100_3way_s51n	58.374	58.364	58.057
58.065 42.200 42.202 41.599			
r10_smoke2sf_lr005_100ep_3way_s42n	59.463	59.423	59.463
59.423 43.730 43.728 43.728			
r10_smoke2sf_lr005_100ep_3way_s43n	60.186	60.171	59.090
59.029 43.730 43.731 43.372			
r10_smoke2sf_lr005_100ep_3way_s44n	60.151	60.142	59.578
59.620 43.190 43.191 43.608			
r10_smoke2sf_lr005_100ep_3way_s45n	60.097	60.088	60.019
60.031 43.170 43.174 43.502			
r10_smoke2sf_lr005_100ep_3way_s46n	59.114	59.131	58.735
58.744 43.790 43.785 43.363			
r10_smoke2sf_lr005_100ep_3way_s47n	59.793	59.749	58.790
58.852 43.710 43.714 43.871			
r10_smoke2sf_lr005_100ep_3way_s48n	59.165	59.202	58.739
58.682 43.390 43.389 43.730			
r10_smoke2sf_lr005_100ep_3way_s49n	59.876	59.875	58.560
58.577 43.500 43.503 43.721			
r10_smoke2sf_lr005_100ep_3way_s50n	59.144	59.108	58.178
58.267 42.990 42.989 43.841			
r10_smoke2sf_lr005_100ep_3way_s51n	59.866	59.892	59.649
59.583 43.760 43.759 43.800			

=====

=====

(1) Verification: can this recomputation reproduce the existing archive

=====

=====

(best,test) recomputed vs sio archive: n=41 max diff 0.0047 median 0.0021 pp

⇒  digit-for-digit agreement (<0.05 pp): the evaluation chain follows the project convention, and the archive values can be reproduced independently

(best,val) recomputed vs results.csv maximum: n=41 max diff 0.0834 median 0.0148 pp

⇒ **i** Exact equality is not expected: Ultralytics selects best.pt by **fitness**, which is not necessarily the maximum of mAP50-95
 (last,val) recomputed vs results.csv final epoch : n=41 max diff 0.0963 median 0.0215 pp

=====
 =====
 (3) Decomposability of Δ gap
 =====
 =====

Identity: $\text{gap} = M_{\text{val}}(\hat{e}) - M_{\text{test}}(\hat{e}) = (V_{\text{final}} - T_{\text{final}}) + (\text{prem_val} - \text{prem_test})$
 $\text{prem_val} = V_{\text{max}} - V_{\text{final}}$ (results.csv); $\text{prem_test} = T_{\text{best}} - T_{\text{final}}$ (measured in this run)

cell	arm	n	V_fin	T_fin	prem_val	prem_test	gap
check							
aitod20	base	1	13.835	10.587	0.786	-0.377	4.411
4.411							
shwd2sf	base	10	59.200	42.824	1.469	0.735	17.110
17.110							
shwd2sf	strat	10	59.532	43.915	0.865	0.151	16.331
16.331							
smoke2sf	base	10	58.022	41.555	0.691	0.507	16.651
16.651							
smoke2sf	strat	10	59.080	43.654	0.605	-0.158	16.189
16.190							

cell	n	Δ gap	Δ diffgap	Δ optimism	sum(check)	dominant
p(Δ gap)						
shwd2sf	10	+0.779	+0.759	+0.020	+0.779	Δ diffgap
0.005258						
smoke2sf	10	+0.461	+1.041	-0.580	+0.461	Δ diffgap
0.02627						

Reading:
 shwd2sf Δ gap=+0.779 (p=0.005258, n=10) = difficulty gap +0.759 (± 1.051) + selection optimism +0.020 (± 0.751)
 smoke2sf Δ gap=+0.461 (p=0.02627, n=10) = difficulty gap +1.041 (± 0.626) + selection optimism -0.580 (± 0.770)

=====

```
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(4) The "realization rate" of the selection gain: how much of the score
picked on val is left on test
=====
=====
```

```
    prem_val = V_max - V_final (book gain obtained by selecting on val)
    prem_test = T_best - T_final (gain realized on test by the same
selection action)
```

cell	arm	n	prem_val	prem_test	realized	per-run
prem_test						
aitod20	base	1	0.786	-0.377	-48%	-0.38
shwd2sf	base	10	1.469	0.735	50%	+0.56 +0.93 +0.66
			+0.18 +0.79 +0.51 +0.61 +1.08 +1.03 +0.99			
shwd2sf	strat	10	0.865	0.151	18%	+0.69 +0.09 +0.31
			+0.20 +0.11 +0.03 -0.09 -0.23 -0.30 +0.71			
smoke2sf	base	10	0.691	0.507	73%	+0.44 +0.59 +0.91
			+0.20 +0.64 +0.66 +0.96 -0.12 +0.19 +0.60			
smoke2sf	strat	10	0.605	-0.158	-26%	+0.00 +0.36 -0.42
			-0.33 +0.43 -0.16 -0.34 -0.22 -0.85 -0.04			

Total: prem_val mean +0.883 pp, prem_test mean +0.172 pp, realization rate 19%

Cells with **negative** prem_test: 2/5 — aitod20/base(-0.377), smoke2sf/strat(-0.158)

⇒ The gain selected on val is only partly realized on an independent split, and can even reverse.

S5 · Selection premium on public third-party logs (sample and reading)

Purpose: main text §5.5; the epoch count, final epoch, best epoch and premium of every log, plus four limitations.

Source: selection_premium_public_20260916.md (md5 04950c92d1ab4bcfb5bbe616b2dcbec4, 9479 bytes); the content below is **embedded verbatim**, unchanged.

Sample

File	Epochs	Final epoch	Best epoch (position)	Premium
max-final	Premium	max-mean of last five epochs	Metric	column name
---	---	---	---	---

aakashchalla12345678-yolov8n__unstated__yolov8n.csv	30	0.4861	0.4861
(ep 30) +0.00 pp +0.26 pp metrics/mAP50-95(B)			
adityadgm070-yolov8-fs-grafik-adamw-results6__unstated__yolov8.csv	100	0.4284	0.4284 (ep 100) +0.00 pp +0.54 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-adamw-results7__unstated__yolov8.csv	100	0.4403	0.4494 (ep 74) +0.91 pp +0.79 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-adamw-results8__unstated__yolov8.csv	100	0.4275	0.4511 (ep 30) +2.36 pp +2.55 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-sgd-results1__unstated__yolov8.csv	100	0.3151	0.3167 (ep 97) +0.16 pp +0.17 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-sgd-results2__unstated__yolov8.csv	100	0.4402	0.4459 (ep 65) +0.57 pp +0.58 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-sgd-results3__unstated__yolov8.csv	100	0.4515	0.4648 (ep 53) +1.33 pp +1.33 pp metrics/mAP50-95(B)
adityadgm070-yolov8-fs-grafik-sgd-results4__unstated__yolov8.csv	100	0.4340	0.4458 (ep 84) +1.18 pp +1.20 pp metrics/mAP50-95(B)
ayyappakorla-yolov3-results__unstated__yolov3.csv	200	0.9011	0.9041 (ep 198) +0.31 pp +0.20 pp metrics/mAP_0.5:0.95
dhanduramesh-yolov11s__ppe-css-data__yolo11n.csv	1	0.4430	0.4430 (ep 1) +0.00 pp - metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolo11.csv	145	0.4647	0.4863 (ep 115) +2.16 pp +2.33 pp metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolo12.csv	150	0.4698	0.4853 (ep 132) +1.55 pp +1.57 pp metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolov8.csv	93	0.4467	0.4621 (ep 63) +1.54 pp +1.49 pp metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model1__unstated__yolov8.csv	123	0.6311	0.6385 (ep 98) +0.74 pp +0.62 pp metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model2__unstated__yolov8.csv	131	0.6290	0.6301 (ep 106) +0.10 pp +0.10 pp metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model3__unstated__yolov8.csv	81	0.6293	0.6435 (ep 56) +1.42 pp +1.27 pp metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model4__unstated__yolov8.csv	182	0.6058	0.6071 (ep 157) +0.14 pp +0.15 pp metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model5__unstated__yolov8.csv	109	0.6294	0.6363 (ep 84) +0.70 pp +0.72 pp metrics/mAP50-95(B)
kimmkless-yolodetector__train2-vehicle-5class__yolov8n.csv	100	0.4327	0.4327 (ep 100) +0.00 pp +0.83 pp metrics/mAP50-95(B)
kimmkless-yolodetector__train4-face-expression__yolov8n.csv	5	0.5803	0.5803 (ep 5) +0.00 pp +12.94 pp metrics/mAP50-95(B)
kimmkless-yolodetector__train8-plant-disease__yolov8n.csv	100	0.4745	0.5029 (ep 80) +2.84 pp +2.82 pp metrics/mAP50-95(B)
nhnthanhhj-metrics-result-f1-yololight__unstated__yolo-light-seg.csv	47		

```
0.6521 | 0.7086 (ep 42) | +5.65 pp | +2.42 pp | metrics/mAP50-95(B) |
| nhnthanhj-metrics-result-f1-yolov8n__unstated__yolov8n-seg.csv | 74 | 0.7093
| 0.7300 (ep 54) | +2.07 pp | +2.10 pp | metrics/mAP50-95(B) |
| ottogalt-yolo-result__unstated__yolov8.csv | 20 | 0.9595 | 0.9595 (ep 20) |
+0.00 pp | +0.56 pp | metrics/mAP50-95(B) |
```

> The "best epoch" is read from each log's own **epoch column** (the ep in the table is the value of that column), not from the row number;
 > most logs in this directory start their epoch at 1, while the yolov3 of ayyappakorla starts at 0; both use column values, so they can be compared across rows.

Logs not included in the statistics (too few epochs to speak of "selection"): aakashchalla12345678-yolov8n__unstated__yolov8n.csv (30 epochs), dhanduramesh-yolov11s__ppe-css-data__yolo11n.csv (1 epoch), kimmkless-yolodetector__train4-face-expression__yolov8n.csv (5 epochs), ottogalt-yolo-result__unstated__yolov8.csv (20 epochs)

Reading

- Complete logs (≥ 40 epochs) number **20** in total, coming from **6** source repositories; the median premium is **+1.04 pp**, the maximum **+5.65 pp**, the minimum **+0.00 pp**; **18/20** are positive.
- Distribution: +0.00, +0.00, +0.10, +0.14, +0.16, +0.31, +0.57, +0.70, +0.74, +0.91, +1.18, +1.33, +1.42, +1.54, +1.55, +2.07, +2.16, +2.36, +2.84, +5.65
- **The most important point:** the largest single premium **+5.65 pp** appears in an ordinary public release log, whereas the effects reported in this kind of literature are often only of the order of 0.5–2 pp—that is, **the selection component can be of the same order as, or even larger than, the reported effect**. The median +1.04 pp is already not small. This explains why "on which split the reported value was selected" is not a detail.
- But the sample is still small (20 logs, 6 sources, and most sources do not annotate the dataset/protocol), so it **can only** be written as "this component is present and quantifiable in other people's published logs"; it **cannot** provide a field-level distribution, nor **can** benchmark aliasing be inferred from it.

Verifiability

- **Byte-level verifiable:** the **SHA-256** and **byte count** of every log are given in SHA256SUMS.txt in the same directory (retrieval date 2026-09-16).
 Anyone can check whether "the bytes analysed" and "the bytes in

circulation" agree.

- **Provenance completed:** PROVENANCE.md in the same directory has now been written. All 24 files are matched to **specific source paths** by content hash or unique byte count (8 Kaggle datasets + 1 GitHub repo, pinned at commit 2417aff0457c), with 0 untraced entries. The mapping is not guessed from filename similarity: the script hashes **every entry** in `_work\kg\*.zip` and collides it against the sha256 of the local files. Check: `python work/build_provenance.py`.

- Two things that **still** cannot be derived from this directory (see PROVENANCE.md §4): ① the **exact downloaded version** on the Kaggle side;
② whether the val/test of these benchmarks are in the same directory (the logs do not carry a yaml).

What is still missing (to be stated as well when writing the main text)

1. **Sample size:** 20 logs, 6 sources—enough to show that the component exists in other people's logs and to give its order of magnitude, but not enough to give a field-level distribution; discussing a distribution requires several tens of traceable logs.
2. **Protocol unknown:** public logs carry no yaml, so it is impossible to judge whether the val and test of a run are the same directory;
therefore "the premium exists" and "benchmark aliasing" are two independent propositions and cannot stand in for each other.
3. **Endpoint convention:** the logs give only per-epoch validation values; public repositories usually do not record "which epoch was reported",
so the premium is an **upper-bound** reading (assuming that what was reported is the best).
4. **Column-name convention:** this table accepts both the new Ultralytics spelling `metrics/mAP50-95(B)` and the old YOLOv5/v3 spelling `metrics/mAP_0.5:0.95` (each row marks which one it uses). The two spellings denote the same quantity, but the **column names differ**—
any script that "filters logs by column name" will miss the other branch; that is the purpose of this item: to make the convention difference visible.

S6 · Process protocols (shared tooling; used by both papers, neither claims originality)

Purpose: main text §2.3 states that "this paper and its companion paper share the same set of process protocols, and neither claims originality".

This section **grounds that statement in verifiable names and hashes** (this is how cross-citation rule ⑤b is implemented).

Form decision (this paper's side): these two protocols are **project-internal artifacts**, and **their text is not released with this paper**——they are not public publications,

and they bear no direct relation to this paper's scientific claims. This paper states only that "**the same set of protocols was used**", and gives the **file names and md5**,

so that a verifier can request the same files from the authors. **If a reviewer requires it, they can be supplied together with the revision.**

Protocol (internal artifact)	md5	Role in this paper	Claims originality?
---	---	---	---
the multi-model review checklist (shared protocol)	e4c73a5ca5992b7af6a5a66c84ba4561	Stratified review by external models; among the six self-reported defects in §10, "silent exclusion" and "wrong column taken" were found by that review	No , neither paper claims it
the local/cloud working-method guide (shared protocol)	10d1b3a3e21f018f3b82e4a293242695	Local/cloud working method: scripted remote execution, all-or-nothing patches, audit anchors, a local pointer for every number	No , neither paper claims it

The same convention also applies to the audit-anchor system: the 291 anchors of this paper and the 494 anchors of the companion paper are the **same** shared tooling,

and neither paper presents it as a methodological contribution.

Why the text is not attached: ① both are **Chinese working documents** with no direct relation to this paper's scientific claims;

② attaching them would make "process" read as "methodological contribution", which is exactly what this rule is meant to avoid; ③ giving the md5 is already enough to confirm that "the two papers use the same one".

S7 · Reproduction entry points (anyone can recompute independently)

① Main-text anchors (291 entries; must ALL PASS)

```
python work\audit_anchors_fupaper.py
```

② 2×2 matrix analysis of S4 (reads the artifacts pulled back in this

```
round)
python work\xeval_analyze_20260916.py
```

```
# ③ Mechanism check of §5.6 (reads the D: archive, read-only)
python work\gap_mechanism_20260916.py
```

```
# ④ Integrity of the artifacts pulled back in this round (sha256 + byte
counts)
python work\verify_pulled.py
python work\verify_pulled.py _xframe
```

Convention statement: the source files of S1–S7 are all different and **must not be mixed**—— S1/S2 are a **configuration-level audit** (no datasets are downloaded),

S3 is a **targeted sampling of official configs/scripts** (5 frameworks, not a census), while S4/S5 are **empirical** (the former on our own runs, the latter on other people's logs).

The strength of every statement in the main text is distinguished accordingly.

S8 · Tables moved from the main text

Why they are here: the venue caps submitted manuscripts at 20–35 double-spaced numbered pages, and these evidence tables cost about eight pages in that format. ****No value has been changed****; every table is reproduced here exactly as it appears, and the main text states each table's headline result in prose.

Four of these tables are also printed in the article — main text Tables 1–4, which are S4 (the Tier 1 test), S6 (the Tier 2 test), S11 (the four counting units) and S20 (the per-cell realization) below — because each carries a headline number the reader is asked to check; the article prints them in the same form as here.

Two further tables deliberately stayed in the article and are not reproduced here: the defect table of main text §10, because disclosure belongs in the article rather than in the Supplementary, and the figure/table inventory, which is a navigation aid.

Index. (n = data rows.)

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Two notes spanning S24–S27 (added 2026-09-18). ① The experiment contains

90 runs: three registered pairs $3 \times 2 \times 10 = 60$, the saturated control $2 \times 5 = 10$

(the registered set is 70), and the control appended by amendment $2 \times 10 = 20$ — counted

from the run directories, ten seeds per arm in every cell but the saturated control.

② **Val-side alias bias**, $\Delta \approx \Delta_{\text{reported}} + \bar{d}$ with $\bar{d} = \text{mean}[p_{\text{baseline}} - p_{\text{strategy}}]$

and $p = \text{max_e} - \text{final}$ from each run's own log: T1-a **+0.051** [0.006, 0.099], T1-c

+0.214 [0.086, 0.326], control C **+0.406** [0.313, 0.498], saturated control

+0.961 [0.440, 1.674] ($\Delta + \bar{d} = -0.529$). Its decomposition: baseline arm **2.505 pp**, strategy arm **1.543 pp** — which is where the article's "about two-thirds of that cell's -1.490 pp" comes from. T1-b is excluded by construction — its YAML

separates the keys. Script `work/x10_alias_audit_20260918.py`. **No verdict changes.**

Table S1 — moved from *2.1 The closest work, and where this paper sits*

Table S1. *2.1 The closest work, and where this paper sits* — columns: Work, What it audits, What it measures, Reports a bias direction, Declares a unit.

Work	What it audits	What it measures	Reports a bias direction	Declares a unit
---	---	---	---	---
Reveles-Martínez et al., <i>Technologies</i> 2026 [1]	Image level: near-duplicate video frames contaminating train/val	Controlled ablation: a random split scores 0.040 mAP50-95 higher than a sequence-aware one (0.787 vs 0.747)	X	X (0 hits for "same directory"/"alias"/"checkpoint selection" in the full text)
Ruangsang & Pramkeaw, <i>Drones</i> 2026 [2]	Image level: near-duplicate training images inside the official test split (46.0 % / 90.7 %)	Removing contamination costs 2.8 / 12.4 points mAP@0.5 ; it does report a val-vs-test gap	Yes, but concludes "no systematic direction" (−0.41 to +0.49 points)	X (0 hits for "same directory"/"alias"/"reported split"/"selection bias")
roboflow/rf-detr PR #1329 [4]	Engineering fix	none	—	—
ultralytics issue #25650 [5]	Engineering fix (3 KITTI Eigen test drives were in train)	none	—	—
CVPR 2026 geospatial leakage audit [30]	Image level (geospatial)	detection + de-duplication	X	X
This paper	Configuration level: the release binds <code>val:/test:</code> to one path	Checkpoint selection premium (best – last) + paired difference Δ_{gap}	Yes, in two tiers: the premium is positive in four cells with the baseline arm gaining more; Δ_{gap} 's sign varies by corpus (2 understate / 2 overstate / 9 n.s.)	Yes: four units, with per-row evidence classes

Table S2 — moved from *3. Units: the methodological premise*

Table S2. 3. Units: the methodological premise — columns: Unit, Definition, Controlled by.

Unit	Definition	Controlled by
---	---	---
release	Does the benchmark's own release artifact (a split script, an archive, a YAML) provide an independent held-out test split?	the releasing party
protocol	What does the official protocol (README, paper, website) declare, and can that test split be evaluated locally ?	the releasing party
yolo_dist	How do the val:/test: keys in the generic YOLO distribution package (e.g. an Ultralytics-style directory) read?	a third party
reported	Which split does the number in the literature come from, and is that split independent of selection?	the user

Table S3 — moved from *5. Finding 2: the selection premium — magnitude, direction, and mechanism*

Table S3. 5. Finding 2: the selection premium — magnitude, direction, and mechanism — columns: Cell, Arm, n, best, last, premium.

Cell	Arm	n	best	last	premium
---	---	---	---	---	---
smoke→SFCHD (strong)	baseline	11	42.156	41.357	+0.799
	strategy	11	43.863	43.481	+0.382
SHWD→SFCHD (corroborating)	baseline	11	43.864	42.826	+1.038
	strategy	11	44.428	43.916	+0.512
dota15→aitod (directional probe)	baseline	10	11.802	11.268	+0.534
	strategy	10	15.401	15.139	+0.262

Table S4 — moved from *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus*

Table S4. 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus — columns: Cell, n, premium baseline, premium strategy, paired diff, sd, t, p, 95 % CI.

Cell	n	premium baseline	premium strategy	paired diff	sd	t	p	95 % CI
---	---	---	---	---	---	---	---	---
smoke2sf (strong)	11	+0.799	+0.382	+0.416	0.240	5.74	1.9×10 ⁻⁴	[+0.255, +0.578]
shwd2sf (corroborating)	11	+1.038	+0.512	+0.526	0.207	8.41	7.6×10 ⁻⁶	[+0.386, +0.665]
a2d15 (directional)	10	+0.534	+0.262	+0.272	0.183	4.69	1.1×10 ⁻³	[+0.141, +0.403]

probe)								
dota15 (within-domain)	10	+0.941	+0.492	+0.449	0.196	7.23	4.9×10^{-5}	[+0.309, +0.589]

Table S5 — moved from *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus*

Table S5. *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus* — columns: Cell, best (selectable), last, last5.

Cell	best (selectable)	last	last5
---	---	---	---
smoke→SFCHD	+1.708	+2.124	+2.086
SHWD→SFCHD	+0.564	+1.090	+1.079
dota15→aitod	+3.599	+3.871	+3.775

Table S6 — moved from *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus*

Table S6. *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus* — columns: Corpus, n, Δgap (pp), p, 95 % CI, Reading.

Corpus	n	Δgap (pp)	p	95 % CI	Reading
---	---	---	---	---	---
shwd2sf	10	+0.779	0.0053	[+0.297, +1.261]	val understates
smoke2sf	10	+0.461	0.0263	[+0.068, +0.855]	val understates
p_aitovis	10	−1.226	1.3×10^{-8}	[−1.370, −1.081]	val overstates
p_vistod15	10	−3.315	6.5×10^{-9}	[−3.676, −2.953]	val overstates
other 9 cells	3–10	−2.46 to +13.66	≥ 0.054	all include 0	not significant

Table S7 — moved from *5.5 The same component in third-party public logs (independent evidence)*

Table S7. *5.5 The same component in third-party public logs (independent evidence)* — columns: Quantity, Value.

Quantity	Value
---	---
Complete logs (≥ 40 epochs)	20 / 6 sources
Median premium	+1.04 pp
Max / min	+5.65 / +0.00 pp
Positive	18/20
Distribution	+0.00, +0.00, +0.10, +0.14, +0.16, +0.31, +0.57, +0.70, +0.74, +0.91, +1.18, +1.33, +1.42, +1.54, +1.55, +2.07, +2.16, +2.36, +2.84, +5.65

Table S8 — moved from *5.6 Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data*

Table S8. 5.6 *Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data* — columns: Cell, n, M_val base / strat, M_test base / strat, Δ gap (pp), p.

Cell	n	M_val base / strat	M_test base / strat	Δ gap (pp)	p
---	---	---	---	---	---
shwd2sf	10	60.67 / 60.40	43.56 / 44.07	+0.779	0.0053
smoke2sf	10	58.71 / 59.69	42.06 / 43.50	+0.461	0.0263
p_aitovis	10	11.24 / 12.66	10.21 / 10.40	-1.226	1.3×10^{-8}
p_vistod15	10	16.96 / 23.93	11.63 / 15.26	-3.315	6.5×10^{-9}
aitod20	3	14.51 / 12.13	10.40 / 9.49	+1.481	0.054
d15d15	3	30.57 / 29.09	15.17 / 14.96	+1.278	0.092
mende20	3	73.40 / 68.03	73.08 / 70.17	+2.472	0.099
fire	3	60.81 / 42.76	39.58 / 35.20	+13.658	0.061
p_d15toai	3	12.83 / 12.31	9.59 / 9.86	+0.781	0.32
mask20	3	34.15 / 31.46	65.35 / 64.42	+1.754	0.66
p_masktomende	3	60.36 / 60.44	69.36 / 66.97	-2.463	0.16
dota	3	30.33 / 29.04	17.17 / 16.24	+0.355	0.56
vis	3	14.19 / 13.39	11.64 / 11.01	+0.169	0.49

Table S9 — moved from *5.6 Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data*

Table S9. 5.6 *Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data* — columns: Candidate mechanism, Prediction, Verdict, Refutation / evidence.

Candidate mechanism	Prediction	Verdict	Refutation / evidence						
---	---	---	---						
M1 multiplicative gain	if the strategy improves by a fixed ratio , Δ gap has the same sign as the raw gap	refuted	3 counterexamples: mask20 (gap -31.2 but Δ gap +1.75), p_aitovis (+1.03 but -1.23), p_vistod15 (+5.34 but -3.32)						
M2 ceiling compression	the easier val is (larger \	gap\	), the more the strategy's advantage is compressed, so Δ gap turns negative as \	gap \	grows	refuted	all six cells with \	gap \	> 10 have Δ gap positive (+0.35 to +13.66); overall $r = +0.32$, $p = 0.28$

M3 level determines resolvability	cells with higher mAP are more likely to be significant	not supported	significant cells' baseline val mean 36.83 pp , non-significant 36.79 pp (Welch p = 0.998)						
M4 whether the strategy truly gains on test	only cells where the strategy gains on test can show val understating	refuted	2 counterexample s: p_aitovis and p_vistod15 are better on test yet have negative Δ gap						
M5 difference in val-side selection gain	Δ gap has the same sign as (premium_baseline – premium_strategy)	not supported	only 8 of 13 cells share the sign; r = +0.06, p = 0.84						

Table S10 — moved from *5.6 Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data*

Table S10. 5.6 Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data — columns: Corpus, Δ gap, final-epoch val diff, Δ prem, –test diff, Dominant.

Corpus	Δ gap	final-epoch val diff	Δ prem	–test diff	Dominant
---	---	---	---	---	---
mask20	+1.75	+15.78	–13.10	–0.93	the two large components nearly cancel
fire	+13.66	+15.96	+2.08	–4.39	final-epoch val diff
p_vistod15	–3.315	–7.072	–0.213	+3.971	final-epoch val diff
p_aitovis	–1.226	–1.583	+0.090	+0.267	final-epoch val diff
shwd2sf	+0.78	–0.33	+0.60	+0.51	Δ prem
smoke2sf	+0.46	–1.06	+0.09	+1.43	–test diff
dota	+0.35	+0.75	+0.54	–0.93	–test diff

Table S11 — moved from *6.1 The result under four units*

Table S11. 6.1 The result under four units — columns: Unit, non-independent, of decidable rows, undecidable.

Unit	non-independent	of decidable rows	undecidable
---	---	---	---
release (the benchmark's own release)	10/19 $\approx$ 53 %	10/19	—
protocol (the official protocol)	13/19 $\approx$ 68 %	13/19	—
yolo_dist (the generic YOLO distribution package)	4/19 $\approx$ 21 %	4/15 $\approx$ 27 %	4 rows

reported (the layer actually reported)	12/19 $\approx$ 63 %	12/15 = 80 %	4 rows
↳ of which literal two-key same path (= the 2/19 the original audit reported)	2/19 $\approx$ 11 %	2/19	—
reported – clean	3/19 $\approx$ 16 %	3/15	4 rows

Table S12 — moved from *6.2 "Same directory" has four levels, and the audit reported only the narrowest*

Table S12. 6.2 "Same directory" has four levels, and the audit reported only the narrowest — columns: Level, Rows, Meaning.

Level	Rows	Meaning
---	---	---
L1 distribution package, literal <code>val == test</code>	2	PASCAL VOC, SFCHD
L2 distribution package, <code>train == val</code>	1	SHWD
L3 distribution package clean, but the reported layer is non-independent	9	Objects365, Open Images v7, DOTA v2.0, UAVDT, xView, NWPU VHR-10, MAFA, WIDER FACE, CrowdHuman
L4 the rest	7	

Table S13 — moved from *6.3 We do **not**** reproduce the original audit's two numbers, and we say why***

Table S13. 6.3 We do not reproduce the original audit's two numbers, and we say why — columns: Audit's number, Our recomputation, Reason.

Audit's number	Our recomputation	Reason
---	---	---
14/19 (the reported layer)	12/19	(h-1) removes D-Fire; DIOR and Mendeley are evidence-insufficient at the reported layer
5/19 clean	3/19	the same two rows move to "undecidable" (they were never adjudicated), so the clean group loses two
12/19 (called "release/spec layer")	release 10/19, protocol 13/19	12 is neither: it was obtained by a two-row manual adjustment (removing WIDER FACE and CrowdHuman), not by applying one unit consistently to all 19 rows

Table S14 — moved from *6.5 Across frameworks: is the convention specific to YOLO?*

Table S14. 6.5 Across frameworks: is the convention specific to YOLO? — columns: Framework (pinned commit), Split the reported metric comes from, Also used for selection?, Status of an independent "test".

Framework	Split the reported metric comes from	Also used for selection?	Status of an independent "test"
-----------	--------------------------------------	--------------------------	---------------------------------

(pinned commit)			
---	---	---	---
Ultralytics YOLO [12] (the yolo_dist unit of §6.1)	the YAML's val:	yes (best.pt; §5's "selection premium" is defined on exactly this convention)	the config has a test : key, but 2 of 19 rows bind it to the same path (§6.1)
Detectron 2 [7] a2f4a8771ab7	DATASETS.TEST: ("coco_2017_val",)	no (TEST.EVAL_PERIOD = 0, no periodic evaluation by default)	coco_2017_test exists, but its annotation file is <code>`image_info_test2017.json`</code> — image information only, no ground truth
MMDetection [6] cfd5d3a985b0	CocoMetric(ann_file='annotations/instances_val2017.json'), val_interval=1	no (CheckpointHook(interval=1), the base config has no save_best)	the base config has no test split at all (train and val dataloaders only)
YOLOX [8] 6ddff4824372	val_ann = "instances_val2017.json"	yes (trainer.py tracks self.best_ap and saves the best)	test_ann is used only when `testdev=True`
PaddleDetection [9] b25522a0f4bd	EvalDataset = val2017	not found by default (runtime.yml and tools/train.py have no save_best); user-side enabling is not excluded	<code>`TestDataset`</code> reads an annotation-free `ImageFolder` , with <code>`anno_path`</code> written as <code>`annotations/instances_val2017.json`</code>
DETR [10] 29901c51d7fe	build_dataset(image_set='val')	no (evaluates every epoch but does not track the best)	the PATHS in datasets/coco.py have only train and val, no test key

Table S15 — moved from *7.2 Both conventions become significant only at $n \geq 6$ *

Table S15. 7.2 Both conventions become significant only at $n \geq 6$ — columns: Cell, n , Δ , paired t p , permutation p ($n = 10$ floor 0.001953), seeds positive.

Cell	n	Δ	paired t p	permutation p ($n = 10$ floor 0.001953)	seeds positive
---	---	---	---	---	---
smoke→SFCHD	10	+1.434 pp	8.6×10^{-7}	0.001953 (= the floor)	10/10
SHWD→SFCHD	10	+0.507 pp	0.0094	0.009766	9/10 (one at -0.19)

Table S16 — moved from *7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)*

Table S16. 7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure) — columns: Reading, Δ , p .

Reading	Δ	p
---	---	---
Taking the final-epoch row (A, 67.12)	-1.490	0.0141
Taking the best-epoch row (B, 70.06)	-0.902	0.2931

Table S17 — moved from *8.2 Results: the effect reproduces, but the magnitude falls*

Table S17. 8.2 Results: the effect reproduces, but the magnitude falls — columns: Cell, clean protocol (n = 10), published (old protocol).

Cell	clean protocol (n = 10)	published (old protocol)
---	---	---
smoke→SFCHD	+1.434 pp	+1.78 (3 seeds) / +1.76 (10 seeds)
SHWD→SFCHD	+0.507 pp	+0.51 (3 seeds) / +0.54 (10 seeds)

Table S18 — moved from *8.3 ★ A direct measurement of selection inflation*

Table S18. 8.3 ★ A direct measurement of selection inflation — columns: Cell, n = 6, n = 9, change.

Cell	n = 6	n = 9	change
---	---	---	---
smoke→SFCHD	+1.562	+1.420	-0.14 pp
SHWD→SFCHD	+0.693	+0.492	-0.20 pp

Table S19 — moved from *8.4 The three variance components: the excluded one is the smaller in this sample*

Table S19. 8.4 The three variance components: the excluded one is the smaller in this sample — columns: Source, What was varied, Magnitude.

Source	What was varied	Magnitude
---	---	---
augmentation RNG	3 augmentation seeds	0.246 pp (strategy) / 0.190 pp (baseline)
initialization (source-side retraining from scratch)	3 independent initializations	0.146 pp (gain) / 0.209 pp (absolute endpoint)
second-stage data order	9–10 seeds	0.403 pp (smoke) / 0.515 pp (SHWD)

Table S20 — moved from *8.6 ★ The premium's realization rate: **19 %***

Table S20. 8.6 ★ The premium's realization rate: 19 % — columns: Cell, Arm, n, prem_val, prem_test, realization rate.

Cell	Arm	n	prem_val	prem_test	realization rate
---	---	---	---	---	---
shwd2sf	baseline	10	+1.469	+0.735	50 %
shwd2sf	strategy	10	+0.865	+0.151	18 %
smoke2sf	baseline	10	+0.691	+0.507	73 %
smoke2sf	strategy	10	+0.605	-0.158	-26 %

aitod20	baseline	1	+0.786	-0.377	-48 %
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A theory-driven prediction, tested on the same runs (added later). §13 derives

realization rate = $1 - WC / \text{prem_val}$ with $WC := E[\varepsilon_{\text{sel}} - \varepsilon_{\text{final}}] \geq 0$, the winner's-curse term.

On the archive: ① $WC = \text{prem_val} - \text{prem_test} = +\mathbf{0.463}$ pp per run ($t = 8.34$, $p = 1.8 \times 10^{-12}$;

the realized premium is smaller in **68/81** runs); ② sorting runs by the noise scale $\hat{\sigma}$ of their

val curve gives realization-rate terciles of **65.6 % / 49.8 % / 13.8 %**, Spearman

$\rho = -\mathbf{0.466}$ ($p = 1.3 \times 10^{-5}$, $n = 80$). The stronger form (premium $\approx \sigma \sqrt{2 \ln E}$) was **withdrawn**:

slope 0.026 ± 0.222 , $p = 0.82$, $n = 101$. Scripts: `work/theory_p4_test_20260919.py`,

`work/theory_meta_and_tests_20260919.py`.

Table S21 — moved from *8.7 The per-epoch view: the realization rate is not a constant*

Table S21. 8.7 *The per-epoch view: the realization rate is not a constant* — columns: Cell · arm, epoch, V(e), T(e), $\Delta V(e)$, $\Delta T(e)$.

Cell · arm	epoch	V(e)	T(e)	$\Delta V(e)$	$\Delta T(e)$
---	---	---	---	---	---
shwd2sf · baseline	20 / 40 / 60 / 80	57.62 / 58.57 / 59.37 / 59.79	41.58 / 42.73 / 43.14 / 43.46	-1.58 / -0.63 / +0.17 / +0.59	-1.24 / -0.10 / +0.32 / +0.63
shwd2sf · strategy	20 / 40 / 60 / 80	52.34 / 56.17 / 58.51 / 59.38	37.98 / 41.21 / 42.69 / 43.91	-7.20 / -3.36 / -1.02 / -0.15	-5.94 / -2.71 / -1.22 / -0.00
smoke2sf · baseline	20 / 40 / 60 / 80	53.69 / 56.25 / 57.20 / 58.08	38.91 / 41.04 / 41.68 / 42.10	-4.33 / -1.77 / -0.82 / +0.05	-2.64 / -0.52 / +0.13 / +0.55
smoke2sf · strategy	20 / 40 / 60 / 80	50.99 / 55.15 / 57.38 / 58.46	36.91 / 41.03 / 42.73 / 43.55	-8.09 / -3.94 / -1.70 / -0.62	-6.74 / -2.63 / -0.92 / -0.11

Complete curve (added 2026-09-19). This table keeps the five-epoch sample. The complete

evaluation — every retained checkpoint of the same 41 runs, 806 rows — is archived as

`xeval_perepoch_20260918/matrix_perepoch.csv` (SHA-256 listed) rather than printed, and the two

agree on all **202** overlapping (run, epoch) points to **0.0000 pp**.

Table S22 — moved from *8.7 The per-epoch view: the realization rate is not a constant*

Table S22. 8.7 *The per-epoch view: the realization rate is not a constant* — columns: Group, n, κ , R^2 .

Group	n	κ	R^2
---	---	---	---
All four arm-cells	765	0.708	0.937
Baseline arm only	385	0.600	0.946
Strategy arm only	380	0.767	0.951

Complete curve (added 2026-09-19). These are the values over **every retained checkpoint** (765 (seed, epoch) points, 41 runs, 806 evaluations, 0 failures). The five-epoch grid of the first version gave **0.734 / 0.532 / 0.769** over $n = 161 / 81 / 80$; the overall estimate therefore moves **−0.026** with overlapping intervals, and the arm difference narrows from +0.237 to **+0.168**. Script: `work/x3b_perepoch_corrected_20260918.py`.

Uncertainty (added 2026-09-18). The epochs within a run are not independent observations, so the fit is re-done with run-clustered inference (41 runs):

all four arm-cells $\kappa = \mathbf{0.734}$, **95 % CI [0.683, 0.784]** (sandwich SE 0.025; a run-level bootstrap gives [0.685, 0.784]); baseline arm **[0.456, 0.608]**; strategy arm **[0.712, 0.827]**.

Resampling runs *within* each arm (the two arms are never pooled) puts the arm difference at **+0.237, 95 % CI [0.145, 0.323]**; a within-run fit, which removes run-level differences entirely, gives 0.774. Scripts: `work/x2_kappa_clustered_20260918.py`, `work/x2b_kappa_arms_20260918.py` (both reproduce the published 0.734 / 0.902 / $n = 161$ first).

Table S23 — moved from *8.7 The per-epoch view: the realization rate is not a constant*

Table S23. *8.7 The per-epoch view: the realization rate is not a constant* — columns: Cell, checkpoints evaluated, $\Delta\text{gap} > 0$, $\Delta\text{gap} < 0$, epochs with $\Delta\text{gap} < 0$, Δgap range.

Cell	checkpoints	$\Delta\text{gap} > 0$	$\Delta\text{gap} < 0$	epochs with $\Delta\text{gap} < 0$	Δgap range (pp)
---	---	---	---	---	---
shwd2sf	19	19	0	—	+0.404 to +4.474
smoke2sf	19	16	3	epochs 5, 10, 15	−4.454 to +1.501

Why this table has changed shape (2026-09-19). The first version sampled epochs {20, 40, 60, 80} and reported "positive at all four sampled epochs" for both cells. With **every** checkpoint evaluated, that holds for `shwd2sf` (all 19) but **not** for `smoke2sf`, whose Δgap is negative at epochs 5, 10 and 15 (−4.454 / −1.089 / −0.058 pp) and positive at every checkpoint from epoch 20 on. The sampled grid began at epoch 20. The four sampled values the old table printed (+1.678 / +0.886 / +0.404 / +0.860 and +0.702 / +1.092 / +0.875 / +1.062) are unchanged and remain

correct for those four epochs. Script: work/x3b_pereepoch_corrected_20260918.py.

Table S24 — moved from *9.2 Result: the **criterion is not met** (the third row is the one that passes)*

Table S24. 9.2 Result: the criterion is not met (the third row is the one that passes) — columns: Registered pair, n, Δ (pp), 95 % CI, paired t, permutation p, same sign, $\Delta \geq +0.30$, $p < 0.01$, $\geq 8/10$, unit criterion.

Registered pair	n	Δ (pp)	95 % CI	paired t	permutation p	same sign	$\Delta \geq +0.30$	$p < 0.01$	$\geq 8/10$	unit criterion
---	--	---	---	---	---	---	---	---	---	---
T1-a dota15→aitod	10	+0.147	[−0.036, +0.330]	1.816 (p = 0.1027)	0.123	6+/4−	✗	✗	✗	FAIL
T1-b aitod→visdrone	10	+0.168	[+0.083, +0.253]	4.455 (p = 0.0016)	0.001953	10+/0−	✗	✓	✓	FAIL
T1-c visdrone→dota15	10	+3.727	[+3.534, +3.920]	43.774 (p = 8.5×10^{-12})	0.001953	10+/0−	✓	✓	✓	PASS

Table S25 — moved from *9.2 Result: the **criterion is not met** (the third row is the one that passes)*

Table S25. 9.2 Result: the criterion is not met (the third row is the one that passes) — columns: Question the frozen criterion asks, Measured.

Question the frozen criterion asks	Measured		
---	---		
\	$\Delta \backslash$	< 0.15 pp? (a saturated domain should be ≈ 0)	no (T2 mask→mendeley −1.490 pp)
$p > 0.1$? (should be non-significant)	no (paired t = −4.164, p = 0.0141 , 0+/5−)		
A positive gain $\geq +0.30$ pp with $p < 0.01$? (if present, the narrative collapses)	no		

Table S26 — moved from *9.3 The near-matched control C: the gain does **not** depend on a label-space change*

Table S26. 9.3 The near-matched control C: the gain does not depend on a label-space change — columns: n, Δ (pp), 95 % CI, paired t, permutation p, same sign.

n	Δ (pp)	95 % CI	paired t	permutation p	same sign
---	---	---	---	---	---
10	+0.727	[+0.640, +0.814]	18.854 (p = 1.53×10^{-8})	0.001953 (= the n = 10 floor)	10+/0−

Table S27 — moved from *9.4 ★ What this section has to do with the paper's subject*

Table S27. 9.4 ★ *What this section has to do with the paper's subject* — columns: Registered pair, data YAML, val, test, Split.

Registered pair	data YAML	val	test	Split
---	---	---	---	---
T1-b aitod→visdrone	visdrone_20p.yaml	images/val	images/test	separated ✓
T1-a dota15→aitod	aitod_20p.yaml	images/val	images/val	same directory ✗
T1-c visdrone→dota15	dota15_20p.yaml	images/val	images/val	same directory ✗
T2 mask→mendeley	mende_20p.yaml	images/val	images/val	same directory ✗

Where these configurations live. The four per-pair configs named above are archived under `eval_validity/yamls/`, each copy named after the path it had on its pod (for example `A__root__datasets__AI-TOD_yolo__aitod_20p.yaml`); that directory also carries a `MANIFEST.md` listing all 28 archived configs and, for each, whether `val` and `test` point at the same directory. The table above is a recomputation from those files, not a restatement.

| C dota→dota15 (control) | dota15_20p.yaml | images/val | images/val | **same directory** ✗ |

Table S28 — moved from *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus*

Table S28. 5.1 *(the dota15 escalation, X8)* — columns: seed, premium baseline, premium strategy, paired difference. Ten seeds, from $n = 3$ to $n = 10$. Premium = best-epoch minus final-epoch mAP50-95 on the run's own monitored split; source: the twenty run directories in `x8_dota15_20260918/runs/` (8 of them trained for this round, seeds 43/44 pre-existing).

seed	premium baseline	premium strategy	paired difference
---	---	---	---
42	+0.997	+0.370	+0.627
43	+1.004	+0.183	+0.821
44	+0.841	+0.317	+0.524
45	+0.775	+0.585	+0.190
46	+0.997	+0.740	+0.257
47	+1.027	+0.658	+0.369
48	+0.986	+0.492	+0.494
49	+0.952	+0.407	+0.545
50	+0.941	+0.715	+0.226
51	+0.891	+0.454	+0.437
mean (n = 10)	+0.941 (sd 0.082)	+0.492 (sd 0.182)	+0.449 (sd 0.196, t = 7.23, p = 4.924e-05)

The criterion was fixed before the runs existed: a ten-seed paired difference failing to be positive with $\geq 8/10$ **same-sign** and a **95 % CI excluding zero** would remove `dota15` from the four-cell headline and force the abstract and Tier 1 to be rewritten. Measured: **+0.449 pp**, **10+/-**, CI [**+0.309**, **+0.589**] — met as stated.

Recipe identity. All twenty `args.yaml` files of the cell (the ten new runs plus the two pre-existing seeds, both arms) are identical in every recipe field: same pretrain (`dota15_pretrain.pt`), same data YAML (`dota15_20p.yaml`), 100 epochs, patience 100, batch 32, `imgsz` 640, `save_period` 5, SGD, `lr0` 0.001 (baseline) versus 0.005 (strategy). Old and new runs are therefore poolable, and the seed is recorded in every directory name.

GPU-memory fallback, and its sensitivity. 10 of the 16 newly trained runs printed `CUDA OutOfMemoryError in TaskAlignedAssigner, using CPU` (92 events; 79 of them in `dota15_lr005_100ep_s51n`), while the per-epoch sweep of §8.7 shared the GPU. This is an implementation-level fallback inside one batch, not a change of objective, and it is reported rather than smoothed: dropping each seed pair in turn leaves the criterion satisfied in all ten subsets (worst lower bound **+0.277**), and dropping the two most affected pairs (s51: 79 events, s42: 7) gives **+0.428 pp**, CI [**+0.252**, **+0.605**], **8+/-**. Scripts: `work/x8_dota15_analysis_20260918.py`, `work/x8_oom_counts_20260918.py`, `work/x8_oom_sensitivity_20260918.py`.

Table S29 — moved from *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus*

Table S29. *5.1 (the two `val`-overstating cells, XI)* — columns: seed, Δ gap for `p_aitovis`, Δ gap for `p_vistod15`. Ten seeds per cell, from `n = 3` to `n = 10`. Δ gap = (arm difference on `val`, at each run's own `val`-curve maximum) – (arm difference on `test`, at `best.pt`); paired within seed. Sources: `x1_tier2_20260918/runs/` (40 run records) and `teval.csv` (80 test evaluations, SHA-256 verified against the machine that produced them).

seed	Δ gap <code>p_aitovis</code>	Δ gap <code>p_vistod15</code>
---	---	---
42	-1.127	-3.131
43	-0.843	-4.563

44	-1.358	-3.058
45	-1.495	-2.819
46	-1.267	-3.468
47	-0.960	-3.453
48	-1.280	-2.952
49	-1.341	-3.590
50	-1.404	-2.995
51	-1.182	-3.118
mean (n = 10)	-1.226	-3.315
95 % CI	[-1.370, -1.081]	[-3.676, -2.953]
paired t / p	-19.17 / 1.3e-08	-20.76 / 6.5e-09
sign	0+/10-	0+/10-
sign-flip permutation p	0.001953	0.001953

What the comparison is, exactly. The plan fixed before the runs: 40 training runs (seeds 42–51 $\times$ 2 arms $\times$ 2 cells), paired *and* sign-flip tests, {best, last} $\times$ {val, test}, and a prespecified correction; the strong branch of the pre-declared outcome rule was *"both remaining negative and significant with permutation $p \leq 0.002$ "*, the weak branch being that the abstract's "direction is not single-valued" must be withdrawn or limited. All ten seeds of each cell are negative and the permutation p is $2/2^{10} = \mathbf{0.001953}$, the smallest value attainable at $n = 10$ (§7.1), so the strong branch applies. **Holm across the two targets** (the prespecified correction) multiplies the larger p by 2 and the smaller by 1: 2.6×10^{-8} and 6.5×10^{-9} , both ≤ 0.05 .

The endpoint was identified, not assumed. The archived `sio_b_results.csv` that the published Table 2 used for the $n = 3$ values carries no endpoint convention (§7.3, disclosed defect 3). Our own evaluation of `best.pt` on `test` reproduces its values for the twelve runs it covers to **0.0046 pp** (median 0.0023) — so that archive records **the best checkpoint evaluated on `test`**, and the $n = 3 \rightarrow n = 10$ comparison is like-for-like rather than a change of measurement. Under the same construction restricted to seeds 42–44 the two cells give -1.110 and -3.584 against the published -1.114 and -3.580 .

Both arms and all four quantities. Per cell (baseline / strategy means, $n = 10$): `val-curve` maximum 11.242 / 12.657 and 16.963 / 23.927; `best.pt` on `test` 10.210 / 10.396 and 11.626 / 15.263.

The cells' realization rates (§8.6's quantity, computed the same way) are **82 % / 104 %** and

56 % / 29 %, pooled 55 % — the opposite end from §8.6's 19 %, which is why the article says the realization rate is not a constant across corpora. The exact three-component decomposition of each Δgap is in Tables S10's conventions: for $p_{\text{aitovis}} -1.226 = (-1.583) + (+0.090) - (+0.267)$ and for $p_{\text{vistod15}} -3.315 = (-7.072) + (-0.213) - (+3.971)$; both identities check to the printed digit. Scripts: `work/x1_gap_analysis_20260918.py`, `work/x1_aux_20260918.py`.

Table S31 — moved from *13.1 Five tests on GPU machines, each with its criterion written down first*

Table S31. *13.1 (G2)* — the selection split's **size** is the only thing that changes: two nested validation splits drawn from one verified-disjoint pool (30 and 90 images), the same training set, and an identical test side. Columns: selection split, arm, n, `prem_val`, `prem_test`, WC, realization rate. Source: `g2_valsize_20260919_runs/, g2t_g1p_20260919/g2_test.csv`.

selection split	arm	n	prem_val	prem_test	WC	realization rate
---	---	---	---	---	---	---
small (30)	base100	5	+2.747	+0.646	+2.101	23.5 %
small (30)	lr005_100ep	5	+1.258	+0.064	+1.194	5.1 %
large (90)	base100	5	+2.549	+0.688	+1.861	27.0 %
large (90)	lr005_100ep	5	+1.062	+0.165	+0.897	15.6 %

Pre-declared criteria. Direction: the realization rate rises with the selection split's size — **met** (the two arms move the same way). Magnitude: $\text{WC}_{\text{small}} / \text{WC}_{\text{large}}$ within $[1.4, 2.1]$ ($1/\sqrt{3} \approx 1.73$) — **not met** (measured 1.19), so the $1/\sqrt{n}$ step is a scale estimate, not a law.

Table S32 — moved from *13.1 Five tests on GPU machines, each with its criterion written down first*

Table S32. *13.1 (G3)* — the epoch budget is the only thing that changes ($E = 100$ vs 200 , same corpus, recipe and validation split). Columns: arm, n, premium at $E = 100$, premium at $E = 200$, difference. Source: `g3_epochs_runs/`.

arm	n	premium (E = 50)	premium (E = 100)	premium (E = 200)	premium (E = 400)	ratio 400/100
---	---	---	---	---	---	---
base	5	+0.724	+0.897	+1.187	+1.315	$\times 1.47$
lr005	5	+0.163	+0.468	+0.756	+1.241	$\times 2.65$

Pre-declared criteria (C1–C3, fixed before the runs). The premium rises with E (regression on $\ln E$): slope **+0.3990** (se 0.0592), $p = 9.466\text{e-}08$, four-budget means monotone **True** — **C1 met**. The $\sigma\sqrt{2 \ln E}$ form predicts $\text{premium}(400)/\text{premium}(100) = 1.141$; the measured ratio is **1.873**, outside the band $[0.86, 1.43]$ — **C2 maintains the shape reading**. Shape control: $\hat{e}/E$ $0.833 \rightarrow 0.877$, $|\Delta| = 0.044$ — **C3 ok**. The extension ($E = 50$ and 400) was added for this revision; it uses the same corpus, recipe, seeds and arms as the $E = 100/200$ runs, and only `--epochs` differs.

Secondary criterion C4 (WC on a disjoint readout) — not met, and systematically so. With $\hat{e}$ selected on `val` and read out on the disjoint third C3 (100 images; $C3 \cap \text{val} = \emptyset$ and $C3 \cap \text{train}$

= 0 verified programmatically), $WC = \text{prem_val} - \text{prem_C3}$ is **negative in all 16 readouts** (mean **-0.616 pp**): the peak-to-endpoint gap is *larger* on the small disjoint split than on val. That is consistent with the shape reading above — the premium is largely the split's own curve shape, and a 100-image split has a steeper curve — and it means Proposition 4's inequality **cannot be tested by reading out on a small disjoint curve**. Two limitations are stated with the result: the readout split has 100 images only, and the four early-stopped $E = 400$ runs are excluded, leaving $n = 6$ at that budget.

Table S33 — moved from *13.1 Five tests on GPU machines, each with its criterion written down first*

Table S33. 13.1 (G4) — replication on a third corpus (mende20_3way, 2 arms $\times$ 10 seeds, test side evaluated separately, 40 evaluations). Columns: run, prem_val, prem_test, WC, realization rate. Source: g4_mende3way_runs/, g4_mende3way_20260919_g4_test.csv.

run	prem_val	prem_test	WC	realization rate
---	---	---	---	---
base100 s42n	+8.873	+3.384	+5.489	38.1 %
base100 s43n	+8.309	+2.701	+5.608	32.5 %
base100 s44n	+7.731	+3.783	+3.948	48.9 %
base100 s45n	+7.818	+2.796	+5.022	35.8 %
base100 s46n	+7.257	+2.481	+4.776	34.2 %
base100 s47n	+7.608	+3.666	+3.942	48.2 %
base100 s48n	+7.534	+3.046	+4.488	40.4 %
base100 s49n	+8.185	+4.312	+3.873	52.7 %
base100 s50n	+8.456	+3.834	+4.622	45.3 %
base100 s51n	+7.671	+2.930	+4.741	38.2 %
lr005 100ep s42n	+9.613	+4.893	+4.720	50.9 %
lr005 100ep s43n	+9.450	+3.434	+6.016	36.3 %
lr005 100ep s44n	+6.249	+0.416	+5.833	6.7 %
lr005 100ep s45n	+8.592	+2.205	+6.387	25.7 %
lr005 100ep s46n	+8.528	+1.960	+6.568	23.0 %
lr005 100ep s47n	+9.957	+5.044	+4.913	50.7 %
lr005 100ep s48n	+7.300	+1.684	+5.616	23.1 %
lr005 100ep s49n	+12.210	+4.314	+7.896	35.3 %
lr005 100ep s50n	+7.976	+2.983	+4.993	37.4 %
lr005 100ep s51n	+7.355	+2.563	+4.792	34.8 %

Result. $WC = +5.212$ pp, 95 % bootstrap CI [+4.822, +5.655], positive in **20 of 20** runs — the same direction as the 81 runs of §8.6 and the 40 of §5.1. No pass/fail is set for the rate itself (Proposition 4 bounds only its expectation).

S9 · Figures moved from the main text

Same reason as §S8. Captions are unchanged apart from the S in the figure number.

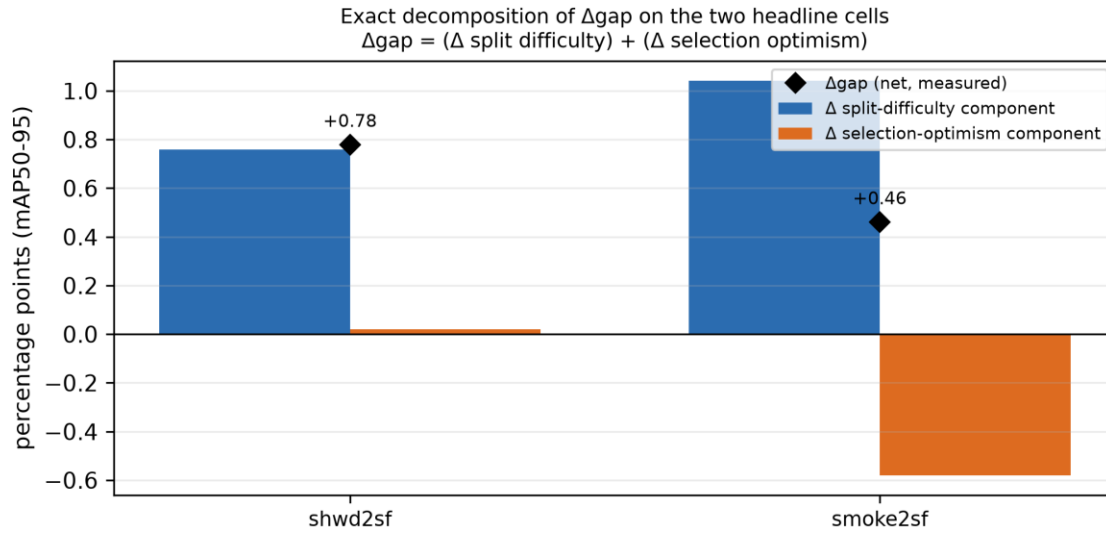


Fig. S1. Exact decomposition of Δgap into a split-difficulty component and a selection-optimism component on the two headline cells. The split-difficulty component dominates (+0.759 of +0.779 for shwd2sf; +1.041 against -0.580 for smoke2sf, where the two components partly cancel). This is an identity, not a fit: the three components sum to Δgap for all 13 cells. Source: `gap_mechanism_20260916.txt`, `xeval_analysis_20260916.txt`.

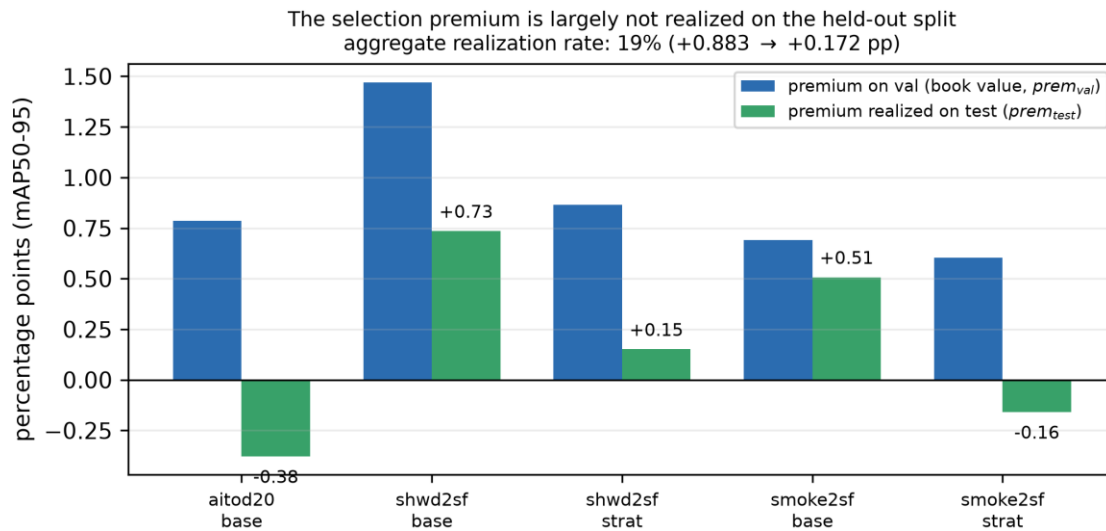


Fig. S2. The premium selected on `val` versus what that same selection action realizes on the disjoint `test` split, per arm-cell (41 three-way runs, 164 evaluations). Aggregate realization rate 19% (+0.883 pp $\rightarrow$ +0.172 pp); in two of five arm-cells the realized premium is negative. Source: `xeval_20260916/matrix.csv`; `xeval_analysis_20260916.txt`.

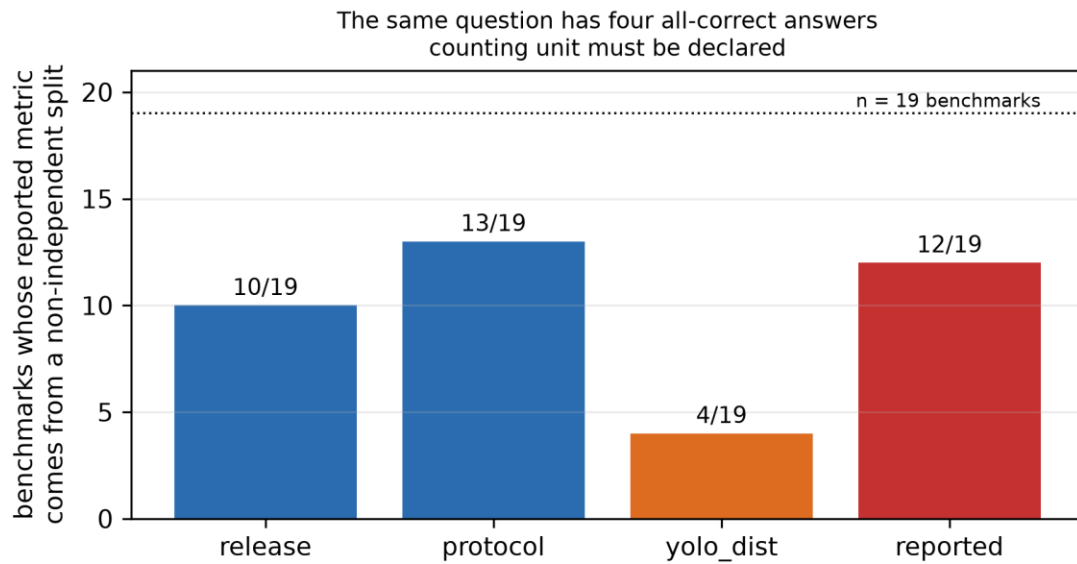


Fig. S3. The same question has four all-correct answers. Under `release` 10/19, under `protocol` 13/19, under `yolo_dist` 4/19 and under `reported` 12/19 benchmarks report a number from a non-independent split ($n = 19$; 4 rows undecidable under two units). A count is meaningless without its unit (main text §3, §6.1). Source: `split_units_recompute_20260916.txt`.

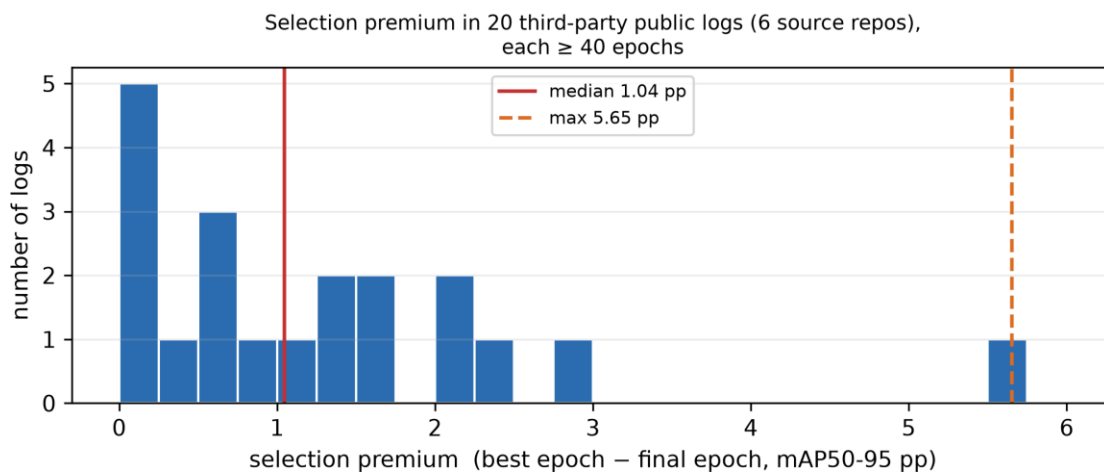


Fig. S4. Selection premium (best – final epoch, mAP50-95 pp) in the 20 complete third-party logs (≥ 40 epochs, 6 source repositories). Median +1.04 pp, maximum +5.65 pp, 18/20 positive, minimum +0.00. The same component measured on our own runs (main text §5.1) is present in logs we did not produce. Source: main text §5.5; `figures/FIGURES.md`.

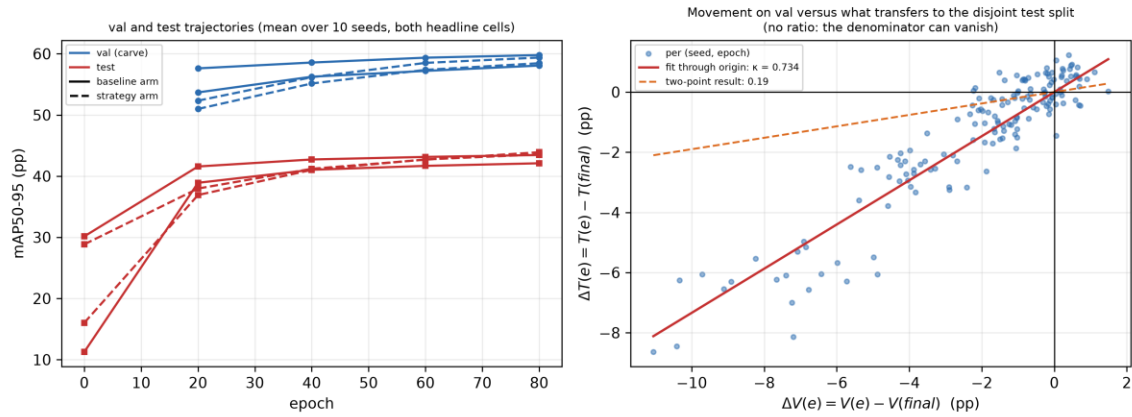


Fig. S5. Per-epoch view of the same runs (41 runs $\times$ 5 intermediate checkpoints, 202 test evaluations). Left: the val and test trajectories. Right: ΔT versus ΔV with the through-origin fit $\kappa = 0.734$ (R^2 0.902, $n = 161$). Source: xeval_20260916/curve.csv; xeval_curve_analysis_20260916.txt.

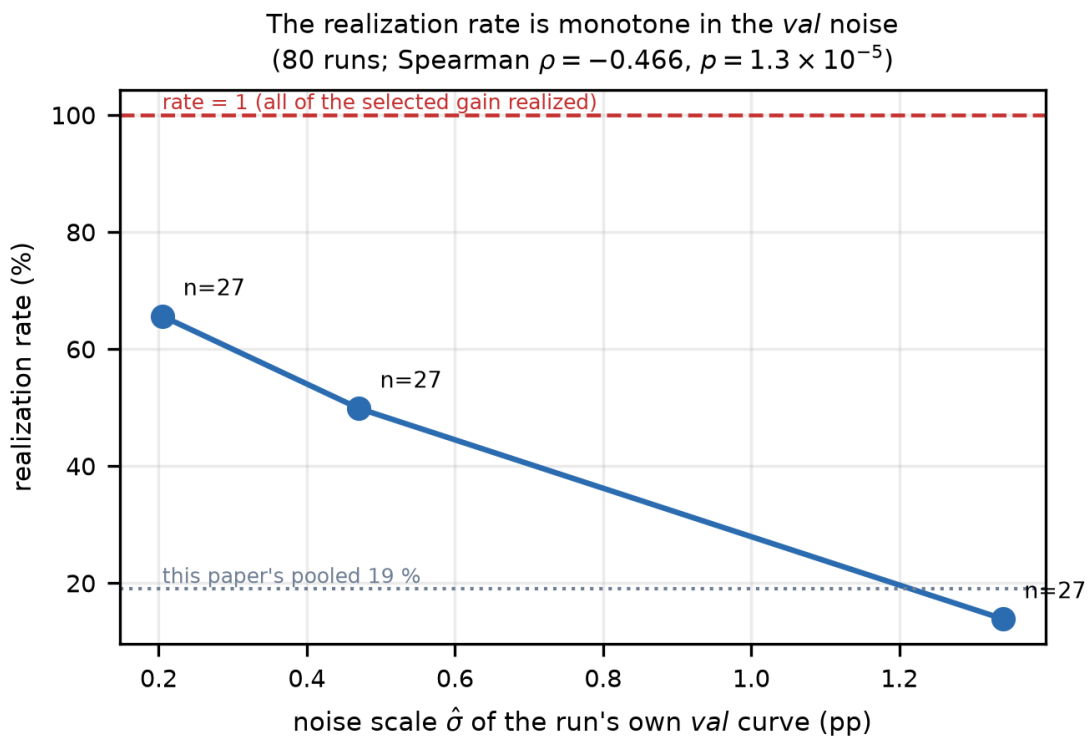


Fig. S6. The realization rate against the noise scale of each run's own val curve — the theory-driven relation of §13's Proposition 4 (rate = 1 – WC / prem_val). Runs are the 80 whose two

sides are both measured (41 three-way runs of §8.2 and the 39 escalated Tier-2 runs of §5.1 whose

test side we evaluated), sorted into terciles by $\hat{\sigma}$, the residual standard deviation of the val

curve around a centred five-point moving average; the realization rate in each tercile is marked with

its n. **Spearman $\rho = -0.466$, $p = 1.3 \times 10^{-5}$** . The dashed line is a rate of 1, i.e. all of the selected gain surviving on test; the dotted line is the pooled 19 % of §8.6. Source:

work/theory_p4_test_20260919.py (figure: work/make_fig6_r54.py).

S11 · Formalisation: propositions and proofs (companion to §13 of the article)

Notation as in §13 of the article: $V(e) = M_{\{S_{\text{val}}\}}(W(e))$, $T(e) = M_{\{S_{\text{test}}\}}(W(e))$, $\hat{e} = \operatorname{argmax}_e V(e)$,

e_{f} = the final checkpoint, $\text{premium}_{\text{val}} = V(\hat{e}) - V(e_{\text{f}})$, $\text{premium}_{\text{test}} = T(\hat{e}) - T(e_{\text{f}})$.

Proposition 1 (non-identifiability of the raw gap)

Statement. $V(\hat{e}) - T(\hat{e}) = [V(e_{\text{f}}) - T(e_{\text{f}})] + [(V(\hat{e}) - V(e_{\text{f}})) - (T(\hat{e}) - T(e_{\text{f}}))]$; hence for any candidate difficulty term δ' the residual $o' = G - \delta'$ reproduces the observation exactly, and no estimator using only the endpoint pair $(\hat{e}, e_{\text{f}})$ identifies δ or o separately.

Proof. Substitute $V(\hat{e}) = V(e_{\text{f}}) + (V(\hat{e}) - V(e_{\text{f}}))$ and $T(\hat{e}) = T(e_{\text{f}}) + (T(\hat{e}) - T(e_{\text{f}}))$; the two expressions are equal identically. For the second claim, fix any $\delta' \in \mathbb{R}$ and set $o' = G - \delta'$; then $\delta' + o' = G$, which is the only equation the data impose. ■

Remark (which assumption buys identifiability). Differencing two arms with matched seeds gives

$\Delta_{\text{gap}} = E[G_{\text{base}} - G_{\text{strat}}] = E[(\delta_{\text{base}} - \delta_{\text{strat}}) + (o_{\text{base}} - o_{\text{strat}})]$. The pairing assumption is

$E[\delta_{\text{base}} - \delta_{\text{strat}}] = 0$, which holds when the two arms are trained and evaluated on one corpus and one

split pair. Under an aliasing configuration both arms' δ are maxima of the *same* split's noise, so the assumption fails by construction: this is the formal content of §9.4.

Proposition 2 (exact three-component identity)

Statement. $\Delta\text{gap} = \Delta\delta_f + \Delta\text{prem_val} - \Delta\text{prem_test}$, where $\Delta\delta_f = (V(e_f) - T(e_f))_{\text{base}} -$

$(V(e_f) - T(e_f))_{\text{strat}}$, $\Delta\text{prem_val} = (\text{premium_val})_{\text{base}} - (\text{premium_val})_{\text{strat}}$, and $\Delta\text{prem_test}$ is the

same for the realized premiums.

Proof. Apply Proposition 1 to each arm and subtract; the split-difficulty terms do not cancel but combine into $\Delta\delta_f$, and the bracket terms are $\Delta\text{prem_val} - \Delta\text{prem_test}$. ■

Verification. The identity holds to the printed digit on all 13 scanned cells (Table S10) and on both escalated cells at $n = 10$: $-1.226 = (-1.583) + (+0.090) - (+0.267)$ and $-3.315 = (-7.072) + (-0.213) - (+3.971)$ (work/x1_aux_20260918.txt).

Proposition 3 (the sign of Δgap does not identify a mechanism)

Statement. If the three components of Proposition 2 are of the same order, then for every target sign $s \in \{+1, -1\}$ there exist admissible component values whose sum has sign s ; hence the observed

sign does not determine which component dominates.

Proof. Let the components be a, b, c with $|a|, |b|, |c| \leq K$. Fixing b and c , choose $a = +2K$ to obtain a positive sum and $a = -2K$ for a negative one; both lie in the allowed range. ■

Empirical support. Measured medians of the three components are **1.07 / 0.54 / 0.93 pp** (same order), the 13-cell scan contains **both signs** (ten positive, three negative; -2.46 to $+13.66$ pp), and candidate mechanisms M1/M2/M4/M5 are each refuted by sign mismatches (Table S9).
**Two

corrections of record.** ① An earlier version of the companion script tried to demonstrate reachability by resampling *cells*; that inherits the cells' own sign split (98.4 % negative on the four cells used) and tests nothing. ② A stronger claim — that the sign is *fragile* — is **not supported: **perturbing one arm-difference term by ± 10 % flips the sign in 0 %**** of cells (± 50 %: 23 %), with a median cancellation ratio of **0.485**. The proposition is therefore stated as "the dominant term varies by corpus and the sign is a sum of same-order terms", which is what the

data support (work/theory_p3a_correct_20260919.txt).

Proposition 4 (realization rate = 1 – winner's-curse share)

Statement. Assume $V(e) = v(e) + \varepsilon_e$ and $T(e) = v(e) + \tau + \eta_e$ with $E[\varepsilon] = E[\eta] = 0$, ε , η

independent of v , and η independent of the selection event $\{\hat{e} = e\}$. Then

$E[\text{premium_test}] = E[\text{premium_val}] - \text{WC}$ with $\text{WC} := E[\varepsilon_{\hat{e}} - \varepsilon_{\{e_f\}}] \geq 0$, so

realization rate $:= E[\text{premium_test}] / E[\text{premium_val}] = 1 - \text{WC} / E[\text{premium_val}] \leq 1$.

Proof. (i) $E[\text{premium_test}] = E[T(\hat{e}) - T(e_f)] = E[v(\hat{e}) - v(e_f)] + E[\eta_{\hat{e}} - \eta_{\{e_f\}}]$; the second term is 0 by the stated independence and mean-zero assumption. (ii) $E[\text{premium_val}] = E[V(\hat{e}) - V(e_f)]$

$= E[v(\hat{e}) - v(e_f)] + E[\varepsilon_{\hat{e}} - \varepsilon_{\{e_f\}}]$. (iii) Subtracting (i) from (ii) yields

$E[\text{premium_test}] = E[\text{premium_val}] - E[\varepsilon_{\hat{e}} - \varepsilon_{\{e_f\}}]$. (iv) Since $\hat{e}$ maximises $V = v + \varepsilon$, the selected

checkpoint carries positive noise in expectation, so $E[\varepsilon_{\hat{e}} - \varepsilon_{\{e_f\}}] \geq 0$; the equality case is

selection on the signal alone. ■

Which assumption carries which half. The inequality ($\leq$) needs only " $\hat{e}$ is chosen using V " — that

is exactly the convention under audit, so the *direction* is robust. The closed form additionally needs η to stay out of the selection; under aliasing (val and test the same split) the two noises coincide and the closed form does not apply — which is why §9.4 treats the aliased cells' realization

figure as a magnitude statement about bias, not as mechanism evidence.

Tests on the archive (work/theory_p4_test_20260919.py, work/theory_meta_and_tests_20260919.py):

test	result
---	---
inequality per run ($\text{WC} > 0$)	+0.463 pp, $t = 8.34$, $p = 1.8 \times 10^{-12}$; realized premium smaller in 68 / 81 runs (sign test $p = 3.8 \times 10^{-10}$)
rate vs val noise (terciles)	65.6 % $\rightarrow$ 49.8 % $\rightarrow$ 13.8 %; Spearman $\rho = -0.466$, $p = 1.3 \times 10^{-5}$, $n = 80$
premium level vs noise ($\sigma \sqrt{2 \ln E}$)	withdrawn : slope 0.026 ± 0.222 , $p = 0.82$, $n = 101$ — the premium's level is set by the curve's shape, not its jitter

Proposition 5 (family dependence of Benjamini–Hochberg)

Statement. For a fixed multiset of p-values and level q , BH's rejection set is determined by the step-up condition on $i/m \cdot q$; consequently the rejection count is not monotone under adding hypotheses:

adding sufficiently small p-values raises the thresholds behind them and can turn a previously non-rejected hypothesis into a rejected one.

Proof. BH rejects $H_{(1..k)}$ for the largest k with $p_{(k)} \leq (k/m)q$. Since $(k/m)q$ is increasing in k , inserting new small p-values shifts existing ones to higher ranks k , which relaxes their thresholds; a hypothesis that failed at rank j can therefore pass at rank $j + t$. ■

Instance in this paper. Adding the two escalated cells' $n = 10$ p-values (6.5×10^{-9} , 1.3×10^{-8}) moved

shwd2sf from "does not survive" to "survives" (0.0053 against a line of 0.01154), and the 13-cell count went from **0 to 3** (work/recompute_bh_20260917.py). Declaring the family is therefore a precondition for the count to be meaningful, not a presentational choice.

Corollary to Proposition 4 (shape v. noise) — derivation and measurements

Derivation (exact). For one arm and one budget E let $v_r(e)$ be run r 's val metric at epoch e ,

$premium_r = \max_e v_r(e) - v_r(E)$, and $m(e) = \max_r v_r(e)$. Put

$shape(E) = \max_e m(e) - m(E)$ and $v_r = premium_r - shape(E)$. Then

$premium_r = shape(E) + v_r$ **by construction**: no assumption about the noise distribution is used.

Two immediate consequences: shape is a functional of the mean curve alone and therefore ****independent**

of σ^{**} ; and $E[v_r] \geq 0$, because replacing each run's curve by the mean curve can only lower the maximum — this is the val-side member of the winner's-curse family of Proposition 4.

Measurement 1 — the premium's level is shape, and the noise term is flat in E (same corpus,

pretraining, two arms, five seeds; only the epoch budget changes; work/prop6_shape_noise_20260920.py):

E	premium (mean of arms)	shape	shape/premium		
---	---------------------------	-------	---------------	--	--

---	---	---	---	---	---
50	+0.444	—	71.2 %		
100	+0.682	—	90.5 %		
200	+0.971	—	87.7 %		
400	+0.907	—	9.2 %		

Four-budget outcome (pre-registered criteria C1–C3). C1 — monotone rise of the premium with E — is **not met**: the series peaks at E = 200 and falls at E = 400 (four of the ten E = 400 runs stopped early under the protocol's own patience, at effective budgets 388/318/362/375; on the six full runs the E = 400 mean is +1.278 pp). C2 — the $\sigma\sqrt{(2 \ln E)}$ form — is **not rejected**: premium(400)/premium(100) = 1.329 against the predicted 1.141, inside the pre-declared band [0.86, 1.43]. C3 — the shape control — **passes** ($\hat{e}/E$ 0.833 $\rightarrow$ 0.815).

E = 50 $\rightarrow$ 200: shape $\times 2.70$, premium $\times 2.19$, $\text{sd}(v) \times 0.92$; the $\sigma\sqrt{(2 \ln E)}$ form would predict $\times 1.24$.

The pre-registered decisive test of that form is criterion C2 (band [0.86, 1.43] on premium(400)/premium(100)); its verdict is reported in §S31–S33 and in §13.1.

Measurement 2 — where the realized gain goes (40 runs with per-epoch test readouts on a 5-epoch

grid, val from each run's own results.csv; selection on val only, readout on test only):

quantity	value		
---	---		
premium(val)	+0.981 pp		
premium(test), oracle peak on test	+0.704 pp (71.7 % of the val premium)		
premium(test), using the val argmax	+0.211 pp (21.5 %)		
runs whose val and test peaks agree	17.5 %		
median \	$\hat{e}_{\text{val}} - \hat{e}_{\text{test}}$		10 epochs (grid step 5)

So the low realization rate is a **peak-alignment** phenomenon, not an absence of gain: an oracle that

could pick the test peak would realize 71.7 % of the val premium, and val-selection retains 30 % of that.

Measurement 3 — selection rules (same 40 runs; every rule chooses on val, readout on test):

rule	realized test premium	captured share	cost to the reported number	within one step of the test peak
---	---	---	---	---
argmax (current convention)	+0.211	30.0 %	0.000	45.0 %

mean of top-2 val epochs	+0.406	57.7 %	0.690	55.0 %
mean of top-3 val epochs	+0.380	54.0 %	0.596	72.5 %
mean of top-5 val epochs	+0.490	69.6 %	0.727	62.5 %
argmax of a 5-epoch moving average	+0.459	65.2 %	0.526	70.0 %
final epoch (no selection)	+0.000	0.0 %	0.981	2.5 %

What these measurements are not. They are a **post-hoc comparison on the same runs**, not a pre-registered experiment: the rules were evaluated after the runs existed. The grid is 5 epochs wide,

so "within one step" means ± 5 epochs; all choices are made on val and every readout is on the other

split; and the oracle row is an upper bound, not an attainable protocol.

A candidate mechanism for the plateau rule — stated as a candidate, not as a result. The natural

explanation of Measurement 3 is that the argmax of a noisy curve is pulled towards epochs whose noise

happens to be high, so its underlying value sits below the true peak (the classical peak-shift penalty,

of order $\sigma^2/|V''|$), while a top-k average estimates the peak *region* with lower variance. That story makes a sharp prediction: the plateau advantage should **grow with the noise of the `val` curve**. We

tested it, and it is **not established**:

* noise estimator: $\sigma \approx$ mean over epochs of the cross-seed standard deviation at that epoch

$(V_r(e) - \text{mean}_r V_r(e))$, i.e. the run-to-run dispersion the ten-seed design is about — not the residual around a smoothing of one run's own curve (which mixes curvature into "noise" and, with that

estimator, gives $\rho = +0.04$, $p = 0.79$);

per run ($n = 40$): $\rho(\hat{\sigma}, \text{plateau} - \text{argmax}) = +0.309$, $p = 0.052^*$;

per cell ($n = 4$ cells, the honest counting unit): $\rho = +0.800$, slope $+0.406 \pm 0.101^*$ per pp of

σ , $r = +0.943$, but **$p = 0.057$** ;

* the four cells read: σ 0.629 $\rightarrow$ +0.194 pp, 0.639 $\rightarrow$ +0.154 pp, 1.125 $\rightarrow$ +0.318 pp, 1.201 $\rightarrow$ +0.447 pp —

i.e. both noisy cells show about twice the advantage of both quiet cells, with no exceptions.

What this is and is not. The *advantage* of the plateau rule is established (Measurement 3, and the

paired test +0.278 pp, $t = 3.678$, $p = 7.1 \times 10^{-4}$, positive in 26 of 40 runs). The *mechanism* is

consistent but not significant on four cells, so we report it as a candidate and not as a result.

Settling it needs more corpus cells with per-epoch test readouts — the direction of the effect is already pinned down by the four we have, but four points cannot carry a significance claim.

13.2 §13.1: the per-comparison reasoning, moved from the article

Where the bias does move, it moves with what the selection sees: the split-size comparison held the test

side fixed and varied only the size (both arms moving the same way), and the epoch-budget comparison left

the peak's relative position essentially unchanged, so neither is a shape artefact. The third-corpus replication evaluated its test side separately and points the same way as the 81 runs of §8.6 and the 40

of §5.1.

Sources: work/g1p_verdict_20260919.py, work/g2_and_g1p_verdict_20260919.txt, work/g1pp_g3_g4_verdicts_20260919.txt, work/theory_p4_test_20260919.py.

8.7b The per-epoch sign detail, moved from the article

shwd2sf's Δ gap is positive at **all 19 checkpoints**; smoke2sf's is ****negative** at the first

three — epochs 5, 10 and 15 — and positive at all 16 checkpoints from epoch 20 onward**;

the four-epoch grid began at epoch 20, so it could not have seen this. At epochs 60 and 80 the val-side

arm difference is $-0.176 / -0.384$ and the test-side one $-1.051 / -1.445$ — ****reproducing the published**

values to the printed digit — **while Δ gap stays positive**: the arm ordering flips on both splits at once, their difference does not.**

13.2b The four-budget detail, moved from the article

Four of the ten $E = 400$ runs stopped early under the protocol's own patience (effective budgets

388/318/362/375); on the six that ran to 400 the $E = 400$ mean is +1.278 pp — and that one restriction

flips both criteria: on the six full runs the series is monotone (C1 met) and

$\text{premium}(400)/\text{premium}(100) = 1.873$ falls *outside* the band (C2 rejected, the shape reading restored).

S12 · Method detail moved from the article

This appendix holds the measurement, protocol and record detail whose **conclusions stay in the article**. It is moved rather than cut: a reader who wants to check a claim reads the same measurements here, while the article keeps one sentence at each site. Identifiers that describe one machine's internal bookkeeping have been replaced by descriptors; the measurements themselves are unchanged.

5.3 The dota15 cell: now reproducible

The local version repaired it for **two reasons**: a stale root snapshot (trees only

up to 2026-09-14; it adds two *_results_20260915 archives), and ****the**

matcher could not see unseeded runs****** — prefix + '_s' matched nothing while

dota15_base100 and dota15_lr005_100ep (**no seed suffix**) exist, so it reported

no runs found forever. The seed suffix is now optional.

It agrees **value by value** with the published readings: three paired best-endpoint

differences **+1.12 / +0.95 / +0.93**, mean **+0.999 $\approx$ +1.00** (last endpoints

+1.77 / +1.46 / +1.54 — **again larger for the baseline arm**, as in §5.1

Tier 1).

A wording correction along the way. M3 wrote this cell as "**3 seeds**", but of the three runs

only two carry a seed (s43, s44); the third (contributing +1.12) has ****no seed identity** in its

directory name******. The values are reproducible, but "3 seeds" is imprecise — it should be disclosed or that run renamed.

Status after the §5.1 escalation (2026-09-19). The cell now exists as **ten seeded runs**

(seeds 42–51: dota15_base100_s*n / dota15_lr005_100ep_s*n). All twenty args.yaml files

—

including the two pre-existing seeds — are identical in every recipe field (same pretrain, same YAML, 100 epochs, batch 32, imgsz 640, SGD, lr0 0.001 vs 0.005), so the old and the new runs are poolable; and the seed now appears in every directory name. ****The wording defect above stays on the record as a defect of the old three-run cell, not of the row now printed.****

9.3 The seed convention: how the ten seeds are realized

Per-run randomness is installed as a **deterministic permutation of the training file order**

(`--shuffle-seed`, set to the number in the run's own name), not through the framework's seed argument: ultralytics 8.4.120's data pipeline does not draw from `args.seed`, and on two of these corpora different `--seed` values produced **identical** epoch-by-epoch replays.

The same convention and the same check hold for the cells escalated in §5.1: all twenty `args.yaml` of the dota15 cell record seed: 42 while its ten per-seed premiums are ten distinct values, and the forty runs of the two escalated Tier-2 cells likewise give ten distinct values in each arm-cell. Script: `work/check_seed_convention_20260918.py`.

5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

Full report: the two-critique verification record; script

`analysis/work/verify_two_critiques_20260916.py` →
`eval_validity/verify_two_critiques_20260916.txt`.

(I) "The premium difference may be noise-level" — refuted. The critique compared it against **0.515 pp**,

the **single-run** data-order component of §8.4, and concluded "underpowered, not significant". The

quantity under test is a **difference of arm means**, with uncertainty the SE of that difference (**0.058–0.120 pp**); the t and p values of §5.1 Tier 1, and their four CIs excluding zero, settle it.

The wrong denominator was used; the data are sufficient.

(II) "You should report $\bar{M}_{val}(\hat{e}) - \bar{M}_{test}(\hat{e})$ directly" — judged non-identifiable. That quantity mixes

the two splits' difficulty with selection optimism and is not identifiable here; we report the ****paired**

difference Δ_{gap} **instead and** state honestly that its sign is not uniform (§5.1 Tier 2). A hard fact that came with it:** in

shwd2sf the arm ordering on val is the reverse of that on test — see §5.1.

What these two checks changed. We used to write "the problem **understates** rather than manufactures a

difference between arms"; (II) corrects this to "***it systematically shifts arm-to-arm comparison, and the

direction varies by corpus**" — the abstract, §2.1, §2.2 and §5.1 rewritten accordingly.

5.5 The same component in third-party public logs (independent evidence)

Sections 5.1–5.4 use **our own** 569 runs; this asks whether the component ****exists** in other people's

published logs**.

Sample. 24 per-epoch training logs; the **20

complete logs (≥ 40 epochs), from 6 source repositories**, are analysed. The definition is identical to ours:

$\text{premium} = \text{best-epoch mAP50-95} - \text{final-epoch mAP50-95}$. Ultralytics-style logs report the

best-validation checkpoint, so this difference is **an upper bound for** the component contributed by selecting on the

reported split.

Table S7 → Supplementary §S8.

What matters most. The largest premium (**+5.65 pp**) occurs in an **ordinary published log** (a 47-epoch

segmentation run; final epoch 0.6521, best epoch 42 at 0.7086). Beside this paper's own magnitudes —

+1.434 pp (§8.2) and a published controlled ablation of **0.040 mAP50-95**

(§2.1) — **the premium's long tail is larger than both**, its median of the same order as the former.

(A magnitude comparison, **not** a same-unit subtraction: a within-run selection difference versus a between-arm effect.)

What it can and cannot support. Can: the component exists in **other people's** published logs, with a

per-log magnitude (all 20 → Supplementary S5).

Cannot: a **field-level** distribution — 20 logs from 6 sources, most not stating dataset or protocol.

The sample is a convenience sample, not a frame: logs came from public dataset hosts and one code repository, filtered only by the inclusion rule above; no platform-side query is named because none was systematic. **Any quotation of these magnitudes** (§2.2, §5.5) carries the four limitations listed below.

Verifiability (upgraded to "byte-checkable *and* traceable"). Each log's ****SHA-256** and byte count **are in `published\_logs/SHA256SUMS.txt`**; **at source level**, 24/24 files** matched to a

source path by **content hash or unique byte count** — 8 Kaggle datasets plus 1 GitHub repository

pinned at commit 2417aff0457c, with **0 untraceable entries** and **24/24 agreement** with SHA256SUMS.txt.

Four limitations:

1. **It cannot establish aliasing**. Public logs carry **no YAML**, so we cannot tell whether that run's `val`

and test share a directory. ****"The premium exists" and "the benchmark aliases" are two independent**

pieces of evidence and cannot substitute for one another** — this section supports only the former (§4, §6 the latter).

2. **The exact downloaded version is unprovable**. What is provable is ****the bytes we analysed equal these**

hashes**", not "these bytes equal the Kaggle version at the time" (the source table records `currentVersionNumber at probe time: ikram0703 2`, the rest 1).

3. **The endpoint reading is an upper bound**: repositories rarely record which epoch was reported, so we

assume it was the best.

4. **Licensing: 5 sources do not state a licence** (Kaggle Unknown), and the GitHub source's `LICENSE`

was not retained. We **do not redistribute** these CSVs, only hashes and source pointers; derived statistics should be cited with that caveat.

A defect we must record (one silent exclusion). The first script version identified metric columns by the

substring 'mAP50-95' in column, recognising **only the new Ultralytics spelling**, so the YOLOv5/v3

spelling metrics/mAP_0.5:0.95 used by ayyappakorla-yolov3-results made read_log return None and the

file **disappeared from the sample with no diagnostic anywhere**. It is a

200-epoch complete log (final epoch 0.9011, best epoch 198 at 0.9041, premium **+0.31 pp**). After the fix

the sample went **19 → 20 logs, 5 → 6 sources**, the median **+1.18 → +1.04 pp**, and positives **17/19 → 18/20**.

Three conventions adopted from that failure (project-wide): ① annotate per row ****which column**

spelling **the log uses**; ② a "logs that could not be parsed must be listed" **section** — a

guard may exclude, but it may not exclude silently; ③ **read the best epoch from each log's own epoch**

column**, not the row index (most start at 1, the ayyappakorla one at 0).

The four logs not analysed run for 30, 20, 5 and 1 epoch; the

5-epoch one's "best – mean of last five" gives a meaningless +12.94 pp — ****which is why a minimum length must**

be imposed**, and we report only max – final, for 20 logs.

Relation to §5.1. This section supplies the existence and magnitude of the same component in ****other**

people's **data**. It does not enter** §5.1's two-tier conclusion (whose evidence is our own paired design), and the two bodies of evidence do **not** corroborate one another.

6.2 "Same directory" has four levels, and the audit reported only the narrowest

Table S12 → Supplementary §S8.

L1 is what the audit enumerated; L3 carries most of the *reported* layer's non-independence.

6.3 We do not reproduce the original audit's two numbers

Table S13 → Supplementary §S8.

6.5 Across frameworks: is the convention specific to YOLO?

Is "the reported metric comes from a non-independent split" a YOLO writing problem, or the default of the

field? §6.1–§6.3 rest on the **generic YOLO distribution package**.

We retrieved, at **pinned commits**, the official configurations and training scripts of **five pinned repositories — 23 files**,

covering six frameworks in total, since the Ultralytics distribution package is the `yolo_dist`

unit of §6.1 and is supplied separately (SHA-256 and byte counts per file in `_xframe/SHA256SUMS.txt`; pins in

`PINNED_COMMITS.txt`; script `work/analyze_xframe_20260916.py` → `xframe_evidence_20260916.txt`), then

read mechanically which split each framework **reports** on and whether that split is **also** used for checkpoint selection.

Table S14 → Supplementary §S8.

Two conclusions: one widens the problem, one narrows our own claim.

(C-1, widening)** Reporting on `val` is the shared default of all five frameworks, because of annotation availability. *** COCO's test annotations are not public — Detectron2 writes the test split's

annotation file as `image_info_test2017.json` (**image information only**), PaddleDetection reads the

same `val` annotation file through a ground-truth-free `ImageFolder`. This is why the reported-layer count is what it is: where annotations are not public, the default

path is to report on whichever split can be scored locally, and that split is usually also the

selection split. **COCO sits on the boundary** — its test-dev is server-scored and cannot be

selected on, so we count its community-reported number as independent, while a paper's own

ablation numbers are not; §6.1 states both readings and both counts. ******On benchmarks like COCO the

coincidence is not an author's oversight — annotation availability pushes everyone onto the same split**, the default path itself.

******(C-2, narrowing) But "select the checkpoint on that split" is not the shared default — only the YOLO

lineage does it. ****** Ultralytics (best.pt) and YOLOX (best_ap) select by the validation metric;

Detectron2 (EVAL_PERIOD = 0), MMDetection (no save_best), DETR (no best tracking) and

PaddleDetection (none found) **report the final epoch**. §5's selection-premium claim therefore reads:

******"within the pipeline family that selects by the validation metric by default (the YOLO lineage), the

convention systematically shifts arm-to-arm comparison" — not ****** "the default of the field". The

abstract and the §2.1 table claim about YOLO distributions only, consistent with this.

Four limitations of this section (so it is not read as a census): ① we read ****default**

configurations, **which users may change**; ② **only** 5 frameworks / 23 files — fixed-point sampling,

not a census; ③ **the pins are** default-branch heads ****** and will move, hence the SHA-256 and commit per

file; ④ as in §4 and §6, a **configuration-level audit — no dataset was downloaded**.

8.1 The protocol

Split the pool three ways — a training subset, a **carved validation** subset for selection, a **test**

subset for reporting; carved and test are disjoint, so the reported number is no

longer the maximum a selection rule saw. Three requirements make it trustworthy:

carve out of the pool (the test split was never in training), ****verify** all three intersections are

zero (**by filename**), and freeze the endpoint definition in writing ****** — which checkpoint counts as "the"

reported number, whose omission is defect 3 in §10 and the source of the conflict in §7.3.

8.3 A direct measurement of selection inflation

Extending the same cell from $n = 6$ to $n = 9$, the mean falls **systematically**:

Table S18 → Supplementary §S8.

The magnitude is 9 %–29 % of the headline effect. ******The estimand is the change

in the cell mean when the seed count rises from 6 to 9** (two nested counts of the same runs; no new

training), so this measures **seed inflation**; reading it as "how much was selection rather than signal" is an interpretation we do not test here.

8.4 The three variance components: the excluded one is the smaller in this sample

Table S19 → Supplementary §S8.

Reading (favourable to this paper): on the same unit, **initialization (0.146 pp) fluctuates less than

data order (0.515 pp) — the component we excluded is not the dominant one** — so "the reported power is

an optimistic upper bound" tightens to "the excluded initialization component is smaller than the dominant

seed component".

Two limitations must be stated with it: ① only 3 initializations, so the SD estimate is very uncertain (with 2 degrees of freedom the 95 % interval for σ is roughly $[0.5\hat{\sigma}, 6\hat{\sigma}]$); we may write "consistent with", **not** "established". ② That experiment ran on machine A with A-side third-party YAMLS,

the $n = 10$ cells on machine B under the same protocol; cross-machine comparability must be noted.

8.5 Why "same-pipeline pairing" is a necessity, not a habit

Three initializations move the **absolute endpoint** of one configuration by 43.250–43.630 (range

0.38 pp, SD 0.209 pp), comparable to seed noise. **Absolute endpoints are not comparable across source

endpoints; only paired differences within one source endpoint are stable.**

5.6 Why Δ gap's sign varies by corpus — measurement detail

§5.1 Tier 2 reports the **fact**; with the archived quantities

the mechanism is not identifiable, though three candidates are **refuted** and two unsupported.

Data: 13 three-way corpus cells $\times$ 2 arms (paired seeds $n = 3$ or 10). Script

work/gap_mechanism_20260916.py → gap_mechanism_20260916.txt; the data path matches §5.4's and

every cell's Δgap agrees **value by value** with the §5.1 Tier 2 table.

Tables S8–S10 → Supplementary §S8.

First, an identity: $\Delta\text{gap} = (\text{arm difference on val}) - (\text{arm difference on test})$, hence

$\Delta\text{gap} > 0 \Leftrightarrow$ the relative arm difference on `val` is smaller than on `test`. It compares the two splits

only and **points to no generating cause**.

An exact decomposition (an identity explaining the unstable sign — **not** a mechanism).

Because $M_{\text{val}}(\hat{e}) = v_{\text{final}} + \text{premium}_{\text{val}}$, Δgap expands into three components, ****all** computable from the

archive******:

$$\Delta\text{gap} = (\text{arm difference at the } \mathbf{final} \text{ epoch on val}) + (\mathbf{val\text{-}side \text{ selection gain} \text{ difference}}) - (\text{test-side arm difference})$$

This is an **identity, not a fit**: for all 13 cells the "check" column (the sum of the three components)

equals Δgap . The three components' median magnitudes are **1.07 / 0.54 / 0.93 pp — the same order**, so

any one of them can flip Δgap 's sign, consistent with §5.1's instability.

The most informative cell is `mask20`: the baseline leads the strategy by **+15.78 pp** at the ****final**

epoch on val, **but the strategy's** val-side selection gain **is 13.10 pp**** larger; the two nearly

cancel, leaving $\Delta\text{gap} +1.75$ ($p = 0.66$, not significant). ****This cell shows that selection alone**

can absorb an arm difference of order 13 pp** — it supports "the convention changes arm-to-arm comparison"

(§5.1 Tier 1) while showing that Δgap 's sign **cannot** be used to infer any single cause.

What this section claims. Δgap 's sign **varies by corpus** is firm (§5.1). Its generating ****mechanism is**

identifiable in this archive only in part******: separating "the two splits' difficulty" from "selection optimism

(one part on val, one on test)" requires **test metrics at non-selected epochs**, and the archive stores only

the test value of the **selected** checkpoint — supplying it means re-running the test evaluation (see §8.6 and §10, defect 5, for what that closed).

What remains is bounded and stated, not left as "mechanism to be explained".

7.2 Both conventions become significant only at $n \geq 6$ — measurement detail

The clean-protocol seed extension (seeds 42–51) reaches **n = 10**:

Table S15 → Supplementary §S8.

First-report attribution (a convention we hold to). The two Δ values here (+1.434 / +0.507 pp) are the **primary readings of the companion paper**, our budgeted fine-tuning study: this paper **cites them and marks them as such**, it does not claim them as first reports. What this paper reports first is ****the derivation of the three-way protocol, the selection premium (§5), and its consequences (§8.3)****. The companion manuscript is named, and its relation to this paper stated, in the cover letter; it carries no identifier while under review, and is available to the editor and the reviewers on request. A formal citation follows once both papers are accepted.

Seed extension is therefore not optional: it upgrades the conclusion from "significant under the parametric test only" to "significant under both".

7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure) — measurement detail

One run, t2_mask2mende_base100_s42n, is recorded with **two different checkpoints**: one row reports

the **final** epoch (67.12), the other the **best** (70.06).

Table S16 → Supplementary §S8.

We follow B and report both. Not an error in either archive: evidence that ****the project had no written convention for which endpoint constitutes "the" reported number**** — §10's defect 3, the reason §3

insists that a count declare its unit.